

# **Decentralized Iris-Boosted Fuzzy Vault: Zero-Penalty Multimodal Template Protection with Threshold-Distributed Storage**

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by

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# **Dissertation approval for M.Tech.**

This dissertation work entitled “**Decentralized Iris-Boosted Fuzzy Vault: Zero-Penalty Multimodal Template Protection with Threshold-Distributed Storage**” by **Nallaparaju V. Suryanarayana Raju (24CSM2S04)** is approved for the degree of Master of Technology (M.Tech).

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This is to certify that the dissertation work entitled “**Decentralized Iris-Boosted Fuzzy Vault: Zero-Penalty Multimodal Template Protection with Threshold-Distributed Storage**” is a bonafide record of the work carried out by **Nallaparaju V. Suryanarayana Raju (24CSM2S04)**, submitted to the faculty of **Department of Computer Science and Engineering**, in partial fulfilment of the requirements for the award of the degree of **Master of Technology in Computer Science and Information Security** at **National Institute of Technology Warangal** during the academic year **2025–26**.

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## Abstract

Biometric template protection is critical for securing identity systems against privacy breaches, with the fuzzy vault serving as a widely studied cryptographic primitive. However, traditional unimodal vaults struggle with noisy data. While multimodal fusion mitigates this, existing multimodal cryptosystems structurally penalize genuine users by treating the secondary biometric as a mandatory gate for decoding. Furthermore, centralized storage of these vaults leaves them highly susceptible to offline brute-force attacks. To address these vulnerabilities, this paper proposes a novel, decentralized multimodal biometric authentication framework implemented in two phases. In the first phase, we propose the Iris-Boosted Fuzzy Vault (IBFV). Fingerprint minutiae are securely mapped onto elliptic curves, while a stabilized iris key generates bonus polynomial points to conditionally boost the vault. If iris recovery fails, the system gracefully reverts to unimodal decoding, achieving a formally proven zero-penalty guarantee. In the second phase, to eliminate single-server trust, we propose D-IBFV, which applies  $(t,n)$ -Shamir Secret Sharing directly to the vault. The shares are distributed across an InterPlanetary File System (IPFS) network, with their cryptographic hashes anchored on a Blockchain ledger to ensure information-theoretic confidentiality and tamper-evident exact reconstruction. The efficacy of the proposed framework is demonstrated through extensive evaluations on the FVC2002, FVC2004, and CASIA-Iris-Interval datasets. IBFV achieves up to a 61.3% relative Equal Error Rate (EER) reduction, yielding EER values (in %) of 0.000, 0.000, 1.384, and 0.929 for FVC2002 DB1, DB2, DB3, and FVC2004 DB1, respectively. System benchmarks on EVM and Hyperledger Fabric confirm practical scalability, yielding sub-second enrollment ( $<806$  ms).

and verification (<71 ms). Furthermore, comprehensive security analysis validates that the proposed framework fully satisfies the ISO/IEC 24745 criteria for irreversibility, unlinkability, and revocability.

**Keywords:** Biometric template protection, fuzzy vault, elliptic curve cryptography, multimodal biometrics, iris recognition, fingerprint, blockchain, Shamir secret sharing, decentralized identity, ISO/IEC 24745.

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## **List of Abbreviations**

|        |                                                    |
|--------|----------------------------------------------------|
| AES    | Advanced Encryption Standard                       |
| AUC    | Area Under the ROC Curve                           |
| BCH    | Bose–Chaudhuri–Hocquenghem (error-correcting code) |
| BFT    | Byzantine Fault Tolerance                          |
| BTP    | Biometric Template Protection                      |
| CID    | Content Identifier (IPFS)                          |
| CLAHE  | Contrast-Limited Adaptive Histogram Equalization   |
| D-IBFV | Decentralized Iris-Boosted Fuzzy Vault             |
| ECC    | Elliptic Curve Cryptography                        |
| ECDLP  | Elliptic Curve Discrete Logarithm Problem          |
| EER    | Equal Error Rate                                   |
| EVM    | Ethereum Virtual Machine                           |
| FAR    | False Acceptance Rate                              |
| FHD    | Fractional Hamming Distance                        |
| FRR    | False Rejection Rate                               |

|      |                                                |
|------|------------------------------------------------|
| FVC  | Fingerprint Verification Competition           |
| GAR  | Genuine Acceptance Rate                        |
| GCM  | Galois/Counter Mode (AES mode)                 |
| GF   | Galois Field (finite field)                    |
| IBFV | Iris-Boosted Fuzzy Vault                       |
| IPFS | InterPlanetary File System                     |
| ISO  | International Organization for Standardization |
| IEC  | International Electrotechnical Commission      |
| NIR  | Near-Infrared                                  |
| PRNG | Pseudo-Random Number Generator                 |
| PRF  | Pseudo-Random Function                         |
| ROC  | Receiver Operating Characteristic              |
| SHA  | Secure Hash Algorithm                          |
| SSS  | Shamir's Secret Sharing                        |
| TPS  | Transactions Per Second                        |
| VSS  | Verifiable Secret Sharing                      |
| ML   | Machine Learning                               |
| API  | Application Programming Interface              |
| HDFS | Hadoop Distributed File System                 |
| AWS  | Amazon Web Services                            |
| GCP  | Google Cloud Platform                          |
| UI   | User Interface                                 |

# Chapter 1

## Introduction

### 1.1 Background and Motivation

Biometric authentication is now deeply embedded in critical infrastructure. Border control systems, national identity programs, mobile banking, and access management platforms routinely rely on fingerprint, iris, and face recognition to verify who a person is [15]. Unlike passwords or tokens, biometric traits are permanent: a user cannot change their fingerprint the way they change a password. This permanence creates an acute problem when biometric *templates*—the stored representations used for matching—are compromised.

ISO/IEC 24745 [14] codifies three mandatory properties for any *Biometric Template Protection* (BTP) scheme. A protected template must be **irreversible**: it must not be possible to reconstruct the original biometric trait from the stored data. It must be **unlinkable**: templates enrolled at different systems (using different transformation parameters) must not be linkable back to the same individual. And it must be **revocable**: when a template is compromised, it must

be possible to re-enroll with a fresh, unlinkable transform. These three requirements together define the security floor for any system claiming to protect biometric privacy.

The *fuzzy vault* [17] is one of the most extensively studied BTP primitives. It hides a secret polynomial among a large set of chaff points; only a biometric query that matches the enrolled data sufficiently closely can identify enough genuine points to reconstruct the polynomial by Lagrange interpolation [31]. Maurya et al. [23] advanced the state of the art by encoding quantized minutiae as elliptic curve (EC) points over Brainpool P256r1 [20], achieving an Equal Error Rate (EER) as low as 0.005% on FVC2002 DB1 with vault entropy exceeding 7.7 bits per coordinate. Despite this progress, unimodal fingerprint systems remain vulnerable to spoof attacks and performance degrades significantly on challenging data: EER reaches 5.0% on FVC2002 DB3.

Multimodal biometric fusion—combining evidence from fingerprint, iris, face, or other traits—is the standard approach to improving accuracy and spoof resistance [32]. However, integrating multimodal fusion *inside* a template protection scheme introduces a fundamental conflict: the cryptographic binding that protects the fingerprint vault must simultaneously accommodate iris data without compromising either modality’s security or the scheme’s ISO/IEC 24745 compliance.

### 1.1.1 The Gating Penalty: A Structural Problem

Every existing multimodal fuzzy vault scheme solves this integration problem by making the auxiliary modality a *mandatory gate*: both modalities must succeed for the vault to be unlocked [25, 24]. The consequence is immediate. Under AND-fusion, the Genuine Acceptance Rate (GAR) of the combined system is:

$$\text{GAR}_{\text{AND}} = \text{GAR}_{\text{FP}} \times \text{GAR}_{\text{Iris}} \leq \text{GAR}_{\text{FP}} \quad (1.1)$$

Since  $\text{GAR}_{\text{Iris}} < 1$  in any realistic deployment, the combined system is always worse than the fingerprint alone for genuine users. With the iris GAR measured at approximately 62% in this work, over one third of genuine users are rejected by the combined system even when their fingerprint would have accepted them.

This is not a calibration problem or a dataset artefact. It is a structural consequence of the AND-fusion architecture: using the iris as a mandatory gate means that iris failure becomes user rejection, regardless of fingerprint quality. The same structural flaw affects polynomial blinding (Architecture B), dense chaff injection (Architecture C), and AES-256-GCM vault encryption (Architecture D)—all tight-coupling strategies examined in this thesis.

### 1.1.2 The Centralization Problem

Even if the biometric performance problem were solved, a separate infrastructure problem remains. Centralized vault storage is a single point of failure. The Aadhaar breach of 2018 exposed biometric records of over 1.1 billion individuals; the U.S. Office of Personnel Management breach of 2015 exposed 5.6 million fingerprint records. Once a vault is leaked, an attacker can mount offline brute-force attacks without time pressure.

Threshold secret sharing—specifically Shamir’s  $(t, n)$ -scheme [33]—offers an information-theoretic solution: split the vault across  $n$  nodes such that any  $t$  shares reconstruct it exactly but any  $t-1$  or fewer shares reveal zero information. No prior work has applied vault-level secret sharing to a fuzzy vault, nor combined it with a zero-penalty multimodal fusion scheme.

## 1.2 Problem Statement

This thesis addresses two tightly linked problems:

1. **The gating penalty.** All existing multimodal fuzzy vault schemes gate decoding on the auxiliary modality. When the iris fails, the user is rejected even if the fingerprint alone would have succeeded. A scheme is needed where a second modality can help when available but never hurt when unavailable.
2. **Centralized vault storage.** Storing protected biometric vaults on a single server creates a single point of failure. A distributed storage architecture is needed that provides information-theoretic security, Byzantine fault tolerance, and auditability.

## 1.3 Objectives

1. To design a multimodal BTP scheme that uses the iris as an optional booster rather than a mandatory gate, and to formally prove  $EER_{IBFV} \leq EER_{Uni}$  for all configurations.

2. To implement and evaluate the proposed IBFV scheme on four FVC fingerprint databases (FVC2002 DB1–DB3, FVC2004 DB1) paired with CASIA-Iris-Interval chimeric samples, across 960+ parameter configurations.
3. To compare IBFV against four alternative multimodal architectures (AND-fusion, polynomial blinding, dense chaff injection, AES-256-GCM encryption) and characterize why tight coupling structurally degrades performance.
4. To design D-IBFV, a decentralized extension applying  $(t, n)$ -Shamir Secret Sharing to the vault itself, distributing shares on IPFS, and anchoring CIDs on a blockchain, with formal proofs of share confidentiality and exact reconstruction.
5. To validate D-IBFV on Ethereum-compatible (Ganache) and Hyperledger Fabric 2.5 platforms, measuring enrollment latency, verification latency, throughput, storage costs, node failure resilience, and accuracy.
6. To demonstrate full ISO/IEC 24745 compliance: irreversibility ( $\text{ECDLP} \sim 2^{128}$ ), unlinkability ( $D_{\text{sys}} \leq 0.03$ ), and revocability via parameter-based re-enrollment.

## 1.4 Contributions

1. **IBFV: The first zero-penalty multimodal fuzzy vault.** The Iris-Boosted Fuzzy Vault uses the iris as an optional booster by injecting  $K$  iris-derived bonus polynomial evaluation points into the fingerprint vault. The zero-penalty guarantee is formally proven (Theorem 1) and empirically validated across 960+ configurations. All four alternative architectures degrade performance, confirming gating is a structural—not incidental—problem.
2. **Full ISO/IEC 24745 security analysis.** A complete analysis against Ratha et al.’s eight-point threat model [29]: irreversibility ( $\text{ECDLP} \sim 2^{128}$ , iris code recovery  $2^{514}$ ), unlinkability ( $D_{\text{sys}} \leq 0.03$ ), and revocability.
3. **D-IBFV: Threshold-distributed vault storage.** The first scheme to split a fuzzy vault via  $(t, n)$ -Shamir Secret Sharing over  $\text{GF}(257)$ , with formal proofs of share confidentiality and exact reconstruction, and blockchain-anchored integrity.
4. **Multi-platform system benchmarks.** Enrollment under 806 ms, verification under 71 ms, zero-bit-error reconstruction across 30,000 trials,  $O(1)$  verification scaling, and only 0.5 KB on-chain per user.

## 1.5 Thesis Organization

Chapter 2 surveys related work. Chapter 3 presents the proposed architecture in full technical detail. Chapter 4 reports biometric and system experiments. Chapter 5 discusses implications, security, and limitations. Chapter 6 concludes with future directions. Appendix A provides mathematical proofs. Appendix B gives complete algorithm listings.

## Chapter 2

### Related Work

This chapter surveys the research landscape across five areas directly relevant to this thesis: biometric template protection, fuzzy vault schemes, multimodal biometric fusion, iris recognition and fuzzy commitment, and blockchain-based biometric systems. A gap analysis table at the end positions this work against the state of the art.

#### 2.1 Biometric Template Protection

Biometric Template Protection (BTP) approaches fall into two broad categories: cancelable biometrics and biometric cryptosystems.

**Cancelable biometrics** [30] apply intentional, repeatable but non-invertible transformations to raw biometric features before storage. The transformed template is stored; the original features are never saved. If the template is compromised, a different transformation produces a

fresh, unlinkable template from the same biometric. Teoh et al. [37] introduced random multi-space quantization for face and palmprint data. The primary criticism of cancelable biometrics is that the transformation may reduce discriminability.

**Biometric cryptosystems** [16] bind a cryptographic key to biometric data such that the key can only be reconstructed when a query biometric is sufficiently similar to the enrolled one. Dodis et al. [9] formalized this binding as *fuzzy extractors*, providing information-theoretically secure constructions with error-tolerance. The most widely studied instantiations are the fuzzy vault [17] and the fuzzy commitment [12].

Rathgeb and Uhl [31] provide a comprehensive survey of both paradigms, concluding that biometric cryptosystems offer stronger provable security guarantees but face greater challenges with noisy, unaligned data. This thesis works within the biometric cryptosystem paradigm.

## 2.2 Fuzzy Vault Schemes

The fuzzy vault [17] was proposed by Juels and Sudan (2002) as a cryptographic construction for binding a secret to an unordered, fuzzy set. A degree- $k$  secret polynomial  $p(x)$  is evaluated at biometric-derived points; these genuine points are hidden among chaff that do not lie on  $p(x)$ . Decoding requires at least  $k+1$  genuine matches for Lagrange interpolation.

Clancy et al. [6] first applied the fuzzy vault to fingerprint minutiae. Uludag et al. [38] added alignment helper data, achieving EER 4.0% on FVC2002 DB2. Nandakumar et al. [26] improved alignment via high-curvature minutiae, reducing EER to approximately 2%.

Tams [36] demonstrated record-multiplicity attacks: multiple vaults from the same user can be cross-matched to identify genuine points. This underscores the need for enrollment-independent structures that ensure vault points are unlinkable across enrollments—a property provided by the ECDLP-based construction of this thesis.

Maurya et al. [23] substantially advanced the state of the art by replacing  $\text{GF}(2^{16})$  encoding with elliptic curve cryptography over Brainpool P256r1 [20]. Minutiae coordinates are quantized and mapped to EC points via  $P_i = q_i \cdot G$ , a one-way mapping protected by ECDLP with security strength  $\sim 2^{128}$  operations. Their scheme achieves EER as low as 0.005% on FVC2002 DB1 with vault entropy exceeding 7.7 bits per coordinate. This ECC vault is the unimodal baseline for the present work.

## 2.3 Multimodal Biometric Fusion

Multimodal biometrics improves recognition accuracy and spoof resistance by combining evidence from multiple traits or sensors [32, 15]. Score-level fusion is most common in unconstrained systems. Inside a fuzzy vault, however, fusion must be integrated at the feature or vault-construction level because no matching score is ever explicitly computed.

Nandakumar and Jain [25] fused fingerprint and iris within a single fuzzy vault by including genuine points from both modalities. Both modalities must successfully contribute to the polynomial reconstruction, creating the AND-fusion constraint:  $\text{GAR}_{\text{AND}} = \text{GAR}_{\text{FP}} \times \text{GAR}_{\text{Iris}} \leq \text{GAR}_{\text{FP}}$ .

Moon et al. [24] applied the same AND-fusion principle to face and fingerprint, with the face module remaining mandatory. Kelkboom et al. [18] fused multiple fingerprint algorithms within a fuzzy commitment framework but did not address cross-modal fusion with a second biometric trait.

Three additional tight-coupling strategies evaluated in this thesis—polynomial blinding (Architecture B), dense chaff seeding (Architecture C), and AES-256-GCM vault encryption (Architecture D)—all require iris key recovery for decoding and exhibit the same AND-fusion penalty. This thesis is the first to break this structural pattern by treating the iris as a booster: contributing when available, falling back to unimodal when not.

## 2.4 Iris Recognition and Fuzzy Commitment

Daugman [7] established the modern framework for iris recognition. The iris is encoded as a binary iris code using multi-scale 2-D Gabor wavelets, with fractional Hamming distance (FHD) as the matching metric. FHD is near 0.10–0.15 for same-iris pairs and near 0.46 (random) for different-iris pairs, providing wide separation. This makes iris one of the most discriminative biometric traits when captured under near-infrared illumination.

Hao et al. [12] introduced the fuzzy commitment scheme for iris codes, binding a 140-bit cryptographic key to a 2048-bit iris code using error-correcting codes. Up to 25% bit-level error can be corrected. This work established that iris codes can be used reliably in fuzzy cryptosystems with appropriate error-correcting codes.

Hollingsworth et al. [13] identified fragile bits in iris codes—positions near phase boundaries where encoding is unstable—and showed that masking them significantly improves matching accuracy.

In this thesis, a repetition-code fuzzy commitment is applied to the full 131,072-bit iris code from CASIA-Iris-Interval images. Block-wise majority decoding with bit interleaving (distributing spatially correlated errors across blocks) achieves a key recovery GAR of approximately 62% at impostor FAR of 0.10%, with block size  $B_s = 255$  bits.

## 2.5 Blockchain-Based Biometric Systems

The intersection of biometrics and blockchain has attracted growing interest as a potential solution to the centralized storage problem.

Delgado-Mohatar et al. [8] surveyed the opportunities and challenges of storing biometric templates on blockchain, identifying high on-chain storage costs, privacy risks from immutable records, and latency as the key barriers. Their key recommendation—off-chain storage with on-chain hash anchoring—is the architecture adopted by D-IBFV.

Acquah et al. [1] confirmed that off-chain storage with on-chain hash anchoring is the scalable pattern and that on-chain template storage incurs prohibitive gas costs at scale.

Goel et al. [10] secured CNN-based biometric feature extraction via blockchain with Shamir-based key recovery. However, unprotected feature data is stored on-chain, providing process auditability without template-level cryptographic protection.

Lee and Jeong [19] proposed BDAS, a blockchain-based decentralized authentication system that fragments templates across a distributed ledger. BDAS achieves decentralization but without a fuzzy vault construction or formal protection guarantees.

Sharma et al. [35] combined fingerprint and iris via cancelable transforms on blockchain (EER  $\sim 8\%$ ). Sharma and Dwivedi [34] deployed a similar approach on Hyperledger Fabric, reporting sub-second latency, without vault-level splitting.

Yun et al. [39] proposed Bio-Rollup, integrating zk-SNARK auditing with layer-2 blockchain scaling for biometric query privacy, but without cryptographic vault protection for stored templates.

Prakasha et al. [28] surveyed privacy-preserving biometric authentication on blockchain, highlighting the need for stronger template protection, scalable architectures, and formal security proofs.

## 2.6 Summary and Gap Analysis

Table 2.1 summarizes the key characteristics of related approaches. Three gaps are directly addressed by this thesis:

1. **No prior multimodal fuzzy vault is penalty-free.** All existing multimodal schemes gate decoding on the auxiliary modality.
2. **No prior blockchain biometric system applies vault-level secret sharing with formal proofs.** Existing systems store templates as monolithic objects.
3. **No prior system demonstrates zero accuracy degradation from decentralization.** D-IBFV proves and empirically confirms bit-exact reconstruction.

**Table 2.1:** Gap analysis: multimodal and blockchain-based biometric schemes.

| Scheme                          | Aux. Required | Zero-Penalty | Decentralized | Formal Proofs |
|---------------------------------|---------------|--------------|---------------|---------------|
| Nandakumar [25]                 | Yes           | No           | No            | No            |
| Moon [24]                       | Yes           | No           | No            | No            |
| Kelkboom [18]                   | Yes           | No           | No            | No            |
| Arch. A–D (this work, compared) | Yes           | No           | No            | No            |
| Delgado-Mohatar [8]             | —             | —            | Yes           | No            |
| Sharma [35]                     | Yes           | No           | Yes           | No            |
| Lee BDAS [19]                   | —             | —            | Yes           | No            |
| Yun Bio-Rollup [39]             | —             | —            | Yes           | No            |
| <b>IBFV (this thesis)</b>       | <b>No</b>     | <b>Yes</b>   | <b>No</b>     | <b>Yes</b>    |
| <b>D-IBFV (this thesis)</b>     | <b>No</b>     | <b>Yes</b>   | <b>Yes</b>    | <b>Yes</b>    |

## **Chapter 3**

### **Proposed Methodology**

This chapter presents the complete technical description of the proposed system in three parts. Section [3.1](#) reviews the building blocks required to understand the construction. Section [3.2](#) presents the IBFV architecture—enrollment, verification, and the formal zero-penalty theorem. Section [3.3](#) describes the D-IBFV decentralized extension.

## 3.1 Preliminaries

### 3.1.1 Elliptic Curve Cryptography

An elliptic curve  $E$  over a prime field  $\text{GF}(p)$  is defined by:

$$y^2 \equiv x^3 + ax + b \pmod{p} \quad (3.1)$$

The points on  $E$ , together with the point at infinity  $O$ , form an abelian group under the chord-and-tangent law. The security of elliptic curve cryptography rests on the **Elliptic Curve Discrete Logarithm Problem** (ECDLP): given a generator  $G$  and a point  $Q = s \cdot G$ , finding the scalar  $s$  is computationally infeasible. For a 256-bit group, the best known attack (Pollard's rho) requires approximately  $2^{128}$  group operations.

This thesis uses **Brainpool P256r1** [20], a 256-bit curve standardized by the German Federal Office for Information Security (BSI) with prime order  $n$ , cofactor  $h = 1$ , and no known special-purpose attack speedups. A key property exploited in the vault construction is that mapping a minutia  $q_i$  to the curve point  $P_i = q_i \cdot G$  is a one-way function: recovering  $q_i$  from  $P_i.x$  (the  $x$ -coordinate stored in the vault) requires solving ECDLP.

### 3.1.2 ECC-Based Fuzzy Vault

The fuzzy vault [17] binds a secret to a noisy, unordered biometric set  $Q$  by evaluating a degree- $k$  polynomial  $p(x)$  at biometric-derived coordinates and hiding the resulting genuine points among randomly generated chaff that do not lie on  $p(x)$ . Unlocking requires identifying at least  $k+1$  genuine points for Lagrange interpolation.

The ECC vault construction of Maurya et al. [23] is adopted as the baseline:

#### Enrollment (Locking)

1. **Quantization:** Each minutia  $m_i = (x_i, y_i, \theta_i)$  is quantized by factor  $f$ :

$$q_i = \left( \left\lfloor \frac{x_i}{f} \right\rfloor \bmod 64 \right) \parallel \left( \left\lfloor \frac{y_i}{f} \right\rfloor \bmod 64 \right) \parallel \left( \left\lfloor \frac{\theta_i}{f} \right\rfloor \bmod 64 \right) \quad (3.2)$$

producing an 18-bit integer. Duplicate values are removed to yield  $Q = \{q_1, \dots, q_g\}$ .

2. **EC Mapping:** Each  $q_i$  is mapped to a curve point  $P_i = q_i \cdot G$ , protected by ECDLP.
3. **Polynomial Generation:** A random degree- $k$  polynomial  $p(x) = \sum_{j=0}^k a_j x^j$  is generated over  $\text{GF}(p)$  with coefficients  $a_j \in [10, 63]$  (6 bits each). The secret key is  $s = \left( \bigoplus_{j=0}^k \text{bin}_6(a_j) \right) \bmod n$ , and the public key is  $S = s \cdot G$ .
4. **Genuine Points:**  $\mathcal{G} = \{(P_i.x, p(P_i.x))\}_{i=1}^g$ .
5. **Chaff:** 5g points  $(c_j.x, r_j)$  where  $r_j \neq p(c_j.x)$ .
6. **Vault:**  $\mathcal{V} = \text{shuffle}(\mathcal{G} \cup C)$ ; stored as  $(\mathcal{V}, S)$ .

### Verification (Unlocking)

Query minutiae  $\mathcal{M}'$  are quantized to  $\mathcal{Q}'$ ; matching vault x-coordinates  $\mathcal{X}_m$  are found. If  $|\mathcal{X}_m| \geq k+1$ , each  $(k+1)$ -subset is tested by Lagrange interpolation; the vault opens if the reconstructed public key matches  $S$ .

### 3.1.3 Iris Code Extraction

Iris images from CASIA-Iris-Interval (320×280 pixels, near-infrared illumination) are processed through a custom pipeline:

1. **Segmentation:** The pupil boundary is detected via intensity thresholding; the outer iris boundary is located via a radial gradient search. A noise mask excludes the pupil interior, specular reflections, and eyelid occlusions (detected via column-wise variance on the normalized strip).
2. **Normalization:** The annular iris region is unwrapped to a  $64 \times 512$  rectangular strip using Daugman's rubber-sheet model [7].
3. **Encoding:** The strip is enhanced via CLAHE and encoded with 1-D Log-Gabor filters at two wavelengths (18 and 36 pixels,  $\sigma/f = 0.30$ ), producing a phase-quantized binary iris code of  $64 \times 512 \times 2 \times 2 = 131,072$  bits. Fragile bits (near phase boundaries, amplitude below  $0.55 \times$  the median) are masked [13]; mean valid bit coverage is approximately 26%.
4. **Matching:** Iris codes are compared via fractional Hamming distance (FHD) over mutually valid bits, with circular shifts ( $\pm 32$ ) columns to compensate for rotational misalignment.

### 3.1.4 Iris Fuzzy Commitment with Bit Interleaving

A repetition-code fuzzy commitment extracts a stable binary key from an iris code. Let  $\mathbf{c} \in \{0, 1\}^N$  be the  $N$ -bit iris code ( $N = 131,072$ ) and  $B_s$  the block size. The number of blocks is  $n_b = \lfloor N/B_s \rfloor$ , each contributing one key bit.

#### Enrollment

1. Partition the iris code into  $n_b$  blocks using **interleaved** assignment: bit  $j$  belongs to block  $(j \bmod n_b)$ . This distributes spatially correlated noise across blocks rather than concentrating it in contiguous regions.
2. Generate a random  $n_b$ -bit key  $\mathbf{k}$ .
3. Construct repetition codeword  $\mathbf{w}$ : position  $j$  is set to  $k_{j \bmod n_b}$ .
4. Compute and store helper data  $\mathbf{h} = \mathbf{c} \oplus \mathbf{w}$  and key hash  $H(\mathbf{k})$  (SHA-256).

#### Recovery

For each rotation shift  $\delta \in [-32, +32]$ :

1. Compute  $\tilde{\mathbf{c}} = \text{rotate}(\mathbf{c}', \delta)$  and  $\tilde{\mathbf{w}} = \tilde{\mathbf{c}} \oplus \mathbf{h}$ .
2. For each block  $i$ , recover  $\hat{k}_i$  by majority vote over the block's valid positions (masking fragile bits). With  $B_s = 255$ , the error capacity per block is  $(B_s - 1)/2 = 127$  bit errors ( $\sim 49.8\%$  error rate per block). Since genuine iris pairs exhibit FHD  $\approx 0.20$  ( $\sim 51$  errors per 255-bit block on average), majority vote succeeds for most blocks. For key recovery to succeed, all  $n_b = 514$  blocks must decode correctly.
3. If  $H(\hat{\mathbf{k}}) = H(\mathbf{k})$ , recovery succeeds.

The resulting GAR is approximately 62% and impostor FAR is approximately 0.10% at  $B_s = 255$ .

### 3.1.5 Shamir's Secret Sharing

Shamir's  $(t, n)$ -threshold Secret Sharing [33] distributes a secret  $s \in \text{GF}(q)$  among  $n$  parties: evaluate a random degree- $(t-1)$  polynomial  $f(x)$  with  $f(0) = s$  at  $n$  distinct points  $\{(j, f(j))\}_{j=1}^n$ . Any  $t$  shares suffice for exact Lagrange reconstruction of  $s$ ; any  $t-1$  or fewer shares reveal zero information (information-theoretic security, regardless of the adversary's computational power).

In D-IBFV, Shamir sharing is applied independently to each byte of the serialized vault over GF(257)—the smallest prime field that contains all 256 byte values (0–255). Using uint16 encoding, each share is 2× the original byte size.

### **3.1.6 Blockchain and IPFS Infrastructure**

#### **Hyperledger Fabric**

Hyperledger Fabric [2] is a permissioned blockchain with pluggable consensus (Raft, BFT) and chaincode (smart contract) execution. Its execute-order-validate model is suitable for institutional deployments. In D-IBFV, a Go chaincode manages CID storage, lookup, and revocation on a Fabric channel, with Raft consensus and a 0.5-second block timeout.

#### **EVM-Compatible Platforms**

Ethereum [4] and its local development proxy (Ganache) provide permissionless smart-contract execution via Solidity. The same Solidity contract deployed on Ganache is EVM-compatible with Polygon, Arbitrum, and other Layer-2 chains. Ganache’s instant block mining provides a controlled baseline for latency measurements.

#### **IPFS**

IPFS [3] is a content-addressed storage network where each object is identified by its SHA-256-based Content Identifier (CID). Storing a Shamir share on IPFS provides: (a) integrity—any tampering changes the CID; and (b) decentralization—the share can be retrieved from any node that pins it. In D-IBFV, each of the  $n$  Shamir shares is stored as a separate IPFS object; the  $n$  CIDs are anchored on the blockchain.

## 3.2 IBFV Architecture

### 3.2.1 Motivation: The AND-Fusion Penalty

Consider a conventional multimodal fuzzy vault where the iris serves as a mandatory gate. Let  $\text{GAR}_{\text{FP}}$  and  $\text{GAR}_{\text{Iris}}$  denote the genuine acceptance rates of the fingerprint and iris subsystems. Under AND-fusion:

$$\text{GAR}_{\text{AND}} = \text{GAR}_{\text{FP}} \times \text{GAR}_{\text{Iris}} \leq \text{GAR}_{\text{FP}} \quad (3.3)$$

Since  $\text{GAR}_{\text{Iris}} < 1$  in practice (approximately 62% in this work), the AND-fusion combined system always performs worse than the unimodal fingerprint for genuine users. Figure 3.1 illustrates the contrast between AND-fusion (gating) and IBFV (boosting).

The same gating problem affects the three alternative tight-coupling architectures:

- **Architecture B (Polynomial Blinding):** The polynomial is XOR-masked with an iris-derived key; decryption requires iris key recovery.
- **Architecture C (Dense Chaff Seeding):** The iris key seeds a PRNG for 20× chaff inflation; without the iris key, the search space for polynomial reconstruction is 20× larger.
- **Architecture D (AES-256-GCM Encryption):** The vault is AES-256-GCM encrypted with the iris key; decryption is impossible without iris recovery.

All three require the iris for decoding, creating an AND-fusion constraint.

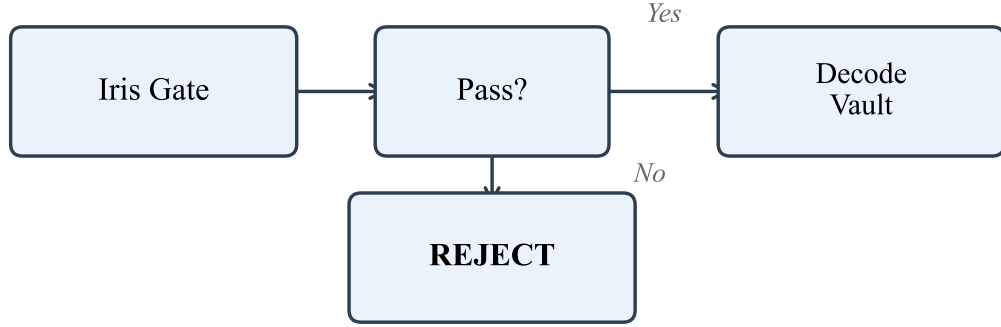
### 3.2.2 IBFV Enrollment

Given fingerprint minutiae  $\mathcal{M}$  and iris code  $(\mathbf{c}, \mathbf{mask})$ , enrollment proceeds as follows (see also Figure 3.2):

1. Execute standard ECC vault enrollment (Section 3.1.2) to obtain genuine points  $\mathcal{G}_{\text{FP}}$ , polynomial  $p(x)$ , and public key  $S$ .
2. Perform iris fuzzy commitment enrollment (Section 3.1.4) to obtain helper data  $(\mathbf{h}, H(\mathbf{k}))$  and key hash  $\kappa = H(\mathbf{k})$ .
3. **Derive virtual x-coordinates:** For  $j = 0, \dots, K-1$ :

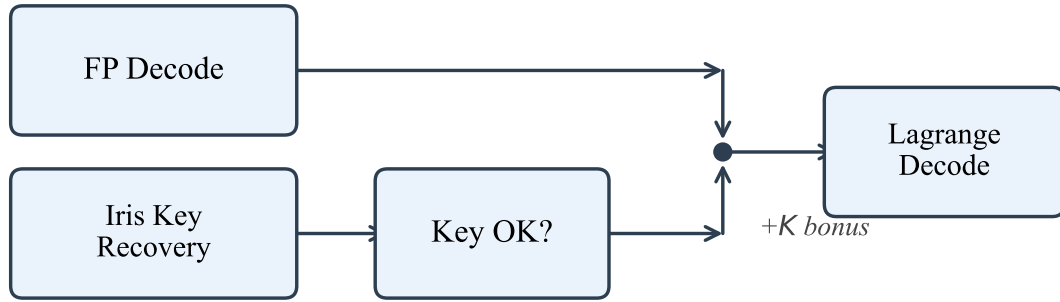
$$v_j = \text{SHA-256}(\text{"ibfv\_vpt:"} \parallel j \parallel \text{"} \parallel \kappa) \bmod p \quad (3.4)$$

**(a) AND-Fusion (Gating)**



$$\text{GAR}_{\text{AND}} = \text{GAR}_{\text{FP}} \times \text{GAR}_{\text{Iris}} \leq \text{GAR}_{\text{FP}}$$

**(b) IBFV (Boosting)**



**Figure 3.1:** AND-fusion vs. IBFV. (a) AND-fusion gates decoding: iris failure blocks the user. (b) IBFV boosts decoding: iris success adds  $K$  bonus points; iris failure falls back to unimodal with zero penalty.

ensuring  $v_j \notin \{P_{i,x}\}$  and all  $v_j$  are distinct.

4. **Compute bonus points:**  $\mathcal{G}_{\text{Iris}} = \{(v_j, p(v_j))\}_{j=0}^{K-1}$ . These are genuine evaluation points of  $p(x)$  seeded by the iris key.
5. **Generate chaff:**  $5|\mathcal{G}_{\text{FP}}|$  chaff points (chaff count based on fingerprint points only, not inflated by bonus points).
6. **Construct vault:**  $\mathcal{V} = \text{shuffle}(\mathcal{G}_{\text{FP}} \cup \mathcal{G}_{\text{Iris}} \cup C)$ .

The stored vault is  $(\mathcal{V}, S, \mathbf{h}, H(\mathbf{k}), B_s)$ .

### 3.2.3 IBFV Verification

Given query fingerprint  $\mathcal{M}'$  and iris code  $(\mathbf{c}', \mathbf{mask}')$ :

1. Quantize query minutiae; find FP-matching vault points:  $\mathcal{X}_{\text{FP}} = \{(x, y) \in \mathcal{V} : x \in \text{EC}(Q')\}$ .
2. Attempt iris key recovery (Section 3.1.4).
3. **If iris succeeds** (key  $\hat{\mathbf{k}}$  recovered):
  - (a) Derive virtual x-coordinates  $\{v_0, \dots, v_{K-1}\}$  using Eq. (3.4).
  - (b) Find bonus matches:  $\mathcal{X}_{\text{Iris}} = \{(x, y) \in \mathcal{V} : x \in \{v_j\}\}$ .
  - (c) Total candidate set:  $\mathcal{X} = \mathcal{X}_{\text{FP}} \cup \mathcal{X}_{\text{Iris}}$ .
  - (d) **Smart decoding**: Fix all  $K$  bonus points (guaranteed genuine) and enumerate only the remaining  $k+1-K$  points from  $\mathcal{X}_{\text{FP}}$ , reducing the search space by  $\approx 5.5\times$  for typical parameters ( $k = 10, K = 4, |\mathcal{X}_{\text{FP}}| = 12$ ).
4. **If iris fails**: Set  $\mathcal{X} = \mathcal{X}_{\text{FP}}$  and proceed with standard Lagrange interpolation—identical to the unimodal vault.

### 3.2.4 Zero-Penalty Theorem

**Theorem 1** (Zero-Penalty Guarantee). *For any parameter configuration  $(f, k)$  and any biometric sample pair  $(\mathcal{M}', \mathbf{c}')$ , the IBFV verification outcome satisfies:*

$$\Pr[\text{Accept}_{\text{IBFV}}] \geq \Pr[\text{Accept}_{\text{Uni}}] \quad (3.5)$$

where the probability is over the randomness in vault construction.

*Proof sketch.* If the iris succeeds, IBFV’s candidate set strictly contains the unimodal set, so any unimodal acceptance also triggers IBFV acceptance. If the iris fails, decoding is identical to unimodal. In neither branch does IBFV reject when the unimodal decoder would accept. Complete proof in Appendix A.

A corollary follows immediately:  $\text{FRR}_{\text{IBFV}} \leq \text{FRR}_{\text{Uni}}$  for all parameter settings. Since the iris impostor FAR is near-zero ( $\approx 0.10\%$  at  $B_s = 255$ ), bonus points are almost never injected for impostors, so  $\text{FAR}_{\text{IBFV}} \approx \text{FAR}_{\text{Uni}}$ , and thus  $\text{EER}_{\text{IBFV}} \leq \text{EER}_{\text{Uni}}$ .

### 3.3 D-IBFV: Decentralized Extension

#### 3.3.1 Motivation

The IBFV vault is a high-value target: server compromise exposes every enrolled user's vault to offline polynomial reconstruction attacks. D-IBFV eliminates single-server trust by applying  $(t, n)$ -Shamir Secret Sharing to the entire vault, storing shares on IPFS, and anchoring their integrity hashes on a blockchain ledger. Figure 3.4 shows the complete architecture.

#### 3.3.2 Vault Serialization

The vault  $(\mathcal{V}, S, \mathbf{h}, H(\mathbf{k}), B_s)$  is deterministically serialized into a byte string  $\mathbf{b} \in \{0, \dots, 255\}^L$ :

- 8-byte header (parameters and format version)
- $|\mathcal{V}| \times 64$  bytes of vault point coordinates
- 64 bytes for the ECC public key  $S$
- $\lceil N/8 \rceil$  bytes of iris helper data  $\mathbf{h}$
- 32 bytes for the key hash  $H(\mathbf{k})$
- 4 bytes of scheme parameters  $(B_s, K, k, f)$
- 32-byte SHA-256 integrity trailer over all preceding bytes

For typical parameters ( $g = 20$ ,  $K = 4$ , chaff =  $5g = 100$  points),  $L \approx 16.6$  KB.

#### 3.3.3 Shamir Vault Splitting

Each byte  $b_i$  of  $\mathbf{b}$  is independently shared over  $\text{GF}(257)$ :

$$f_i(x) = b_i + a_1^{(i)}x + \dots + a_{t-1}^{(i)}x^{t-1} \pmod{257} \quad (3.6)$$

with coefficients  $a_j^{(i)} \xleftarrow{\$} \text{GF}(257)$ . Share  $j$  receives  $(j, f_i(j))$  for each byte  $i$ .

Two theorems govern the security and correctness of this construction:

**Theorem 2** (Share Confidentiality). *For any  $t' < t$  compromised shares, the conditional distribution of vault  $\mathbf{b}$  is uniform over  $\{0, \dots, 255\}^L$ . Fewer than  $t$  shares reveal zero information about the vault content.*

**Theorem 3** (Exact Reconstruction). *Given any  $t$  shares with distinct indices, Lagrange interpolation over  $\text{GF}(257)$  recovers  $\hat{b}_i = b_i$  for all  $i$ , yielding  $\hat{\mathbf{b}} = \mathbf{b}$  with zero bit errors.*

Complete proofs are given in Appendix A.

### 3.3.4 Distributed Enrollment Protocol

The enrollment protocol extends IBFV enrollment with three steps:

1. Run IBFV enrollment to obtain the vault  $(\mathcal{V}, S, \mathbf{h}, \kappa, B_s)$ .
2. Serialize the vault to byte string  $\mathbf{b}$ .
3. Apply  $(t, n)$ -Shamir splitting to obtain  $n$  shares  $\{\mathbf{s}_1, \dots, \mathbf{s}_n\}$ .
4. Upload each share  $\mathbf{s}_j$  to IPFS; record its CID  $\text{cid}_j$ .
5. Store  $(\text{uid}, \{\text{cid}_1, \dots, \text{cid}_n\}, t, n)$  on the blockchain.
6. Securely erase  $\mathbf{b}$  and all shares from local memory.

The complete vault never exists on any server; only individual shares (which are information-theoretically useless without  $t-1$  others) are persisted.

### 3.3.5 Distributed Verification Protocol

1. Retrieve the CID list and threshold  $t$  from the blockchain for the claimed user ID.
2. Download any  $t$  available shares from IPFS; verify each share against its CID (IPFS content addressing provides automatic integrity verification).
3. Reconstruct  $\hat{\mathbf{b}}$  via Lagrange interpolation over  $\text{GF}(257)$ .
4. Verify the SHA-256 integrity trailer embedded in  $\hat{\mathbf{b}}$ .
5. Deserialize the vault from  $\hat{\mathbf{b}}$ .
6. Run IBFV verification against the reconstructed vault.

7. Securely erase  $\hat{\mathbf{b}}$  and all shares from local memory.

The reconstructed vault exists only transiently in memory on the verification client and is erased immediately after the biometric decision.

### 3.3.6 Revocation and Re-Enrollment

To revoke a compromised vault:

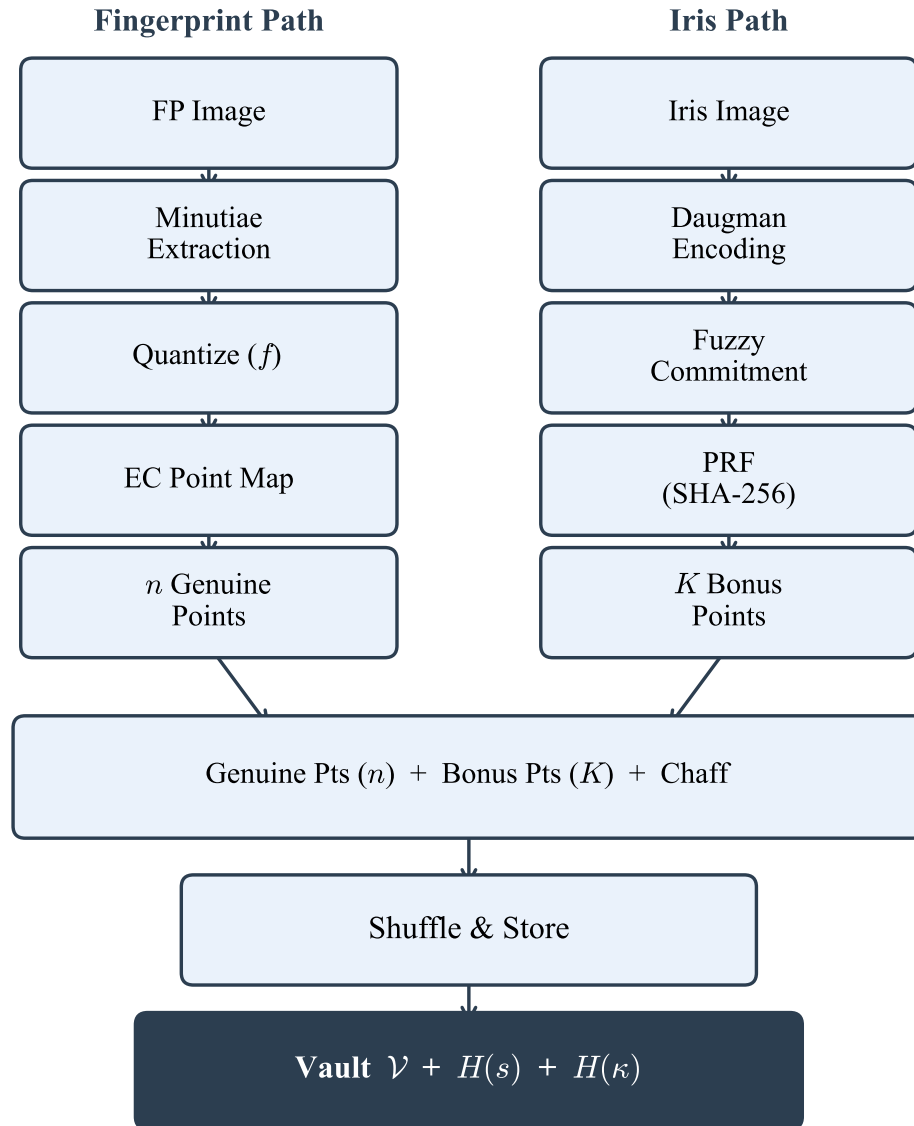
1. Submit a revocation transaction to the blockchain, marking the current CID set as revoked. The blockchain's append-only nature provides a permanent, tamper-evident audit trail.
2. Re-enroll with fresh parameters: a new polynomial, a new iris commitment key, and new Shamir randomness. By Theorem 2, new shares are information-theoretically independent of revoked shares.
3. Upload new shares to IPFS; record new CIDs on the blockchain.

### 3.3.7 Extended Threat Model for D-IBFV

D-IBFV adds three attack points beyond the standard eight-point model:

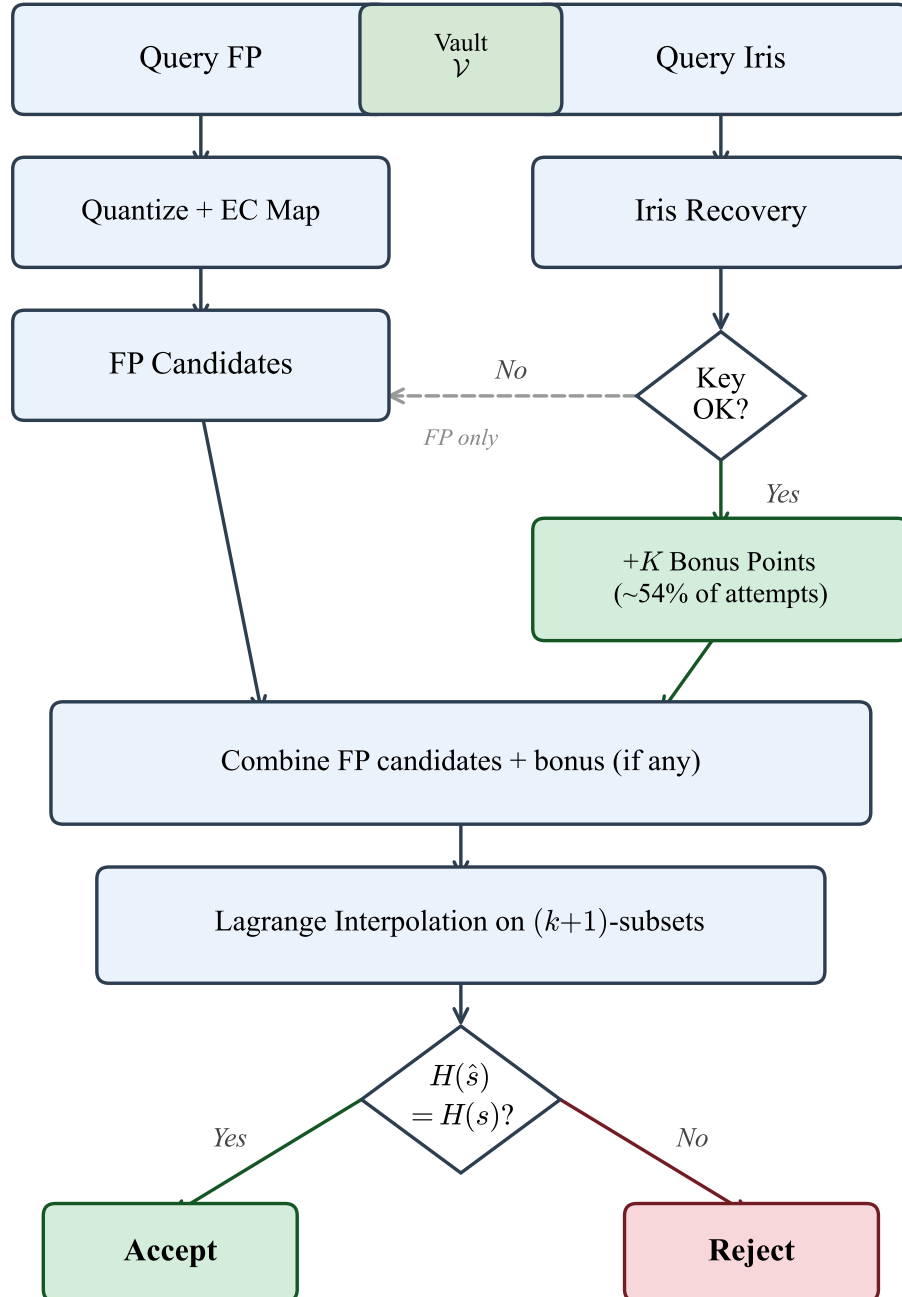
- **Node compromise:** An adversary compromises up to  $t-1$  IPFS storage nodes. Countermeasure: Theorem 2 guarantees zero information leakage from  $t-1$  or fewer shares.
- **Blockchain tampering:** An adversary attempts to modify CID records. Countermeasure: Blockchain consensus (Raft/BFT) makes retroactive modification computationally/information-theoretically infeasible. IPFS CIDs independently verify share integrity.
- **IPFS share substitution:** An adversary replaces a share blob. Countermeasure: IPFS CIDs are cryptographic content hashes; substitution changes the CID. The blockchain-anchored CID and the vault's SHA-256 integrity trailer jointly detect any corruption.

## IBFV Enrollment

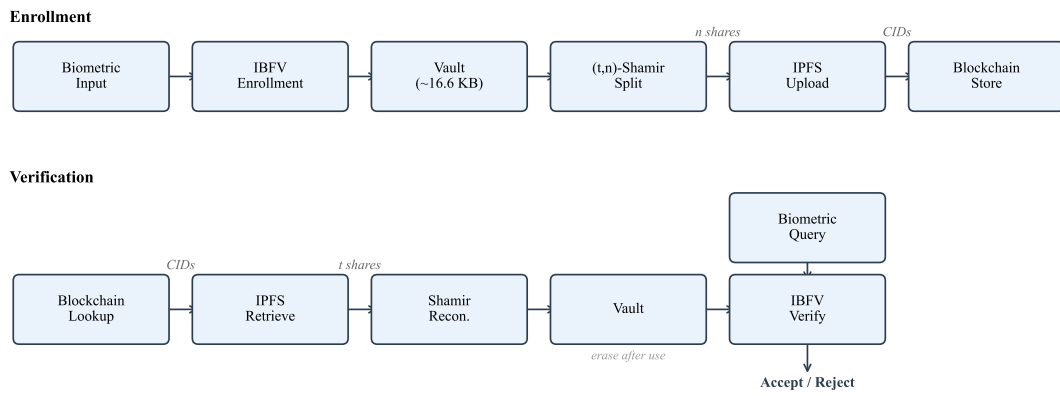


**Figure 3.2:** IBFV enrollment pipeline. The fingerprint path maps minutiae to EC points on Brainpool P256r1. The iris path extracts a key via fuzzy commitment, seeding a PRF to derive  $K$  bonus points. All points are merged with chaff.

### IBFV Verification



**Figure 3.3:** IBFV verification pipeline. If iris recovery succeeds ( $\sim 62\%$ ),  $K$  bonus points reduce the search space. If it fails (dashed path), the system falls back to unimodal decoding with zero penalty.



**Figure 3.4:** Complete D-IBFV architecture. Top: enrollment pipeline. Bottom: verification pipeline. The vault never exists as a complete unit on any single node except transiently on the trusted enrollment/verification client.

## Chapter 4

### Experimental Results

This chapter presents experimental evaluation in two parts: biometric performance experiments (§4.1–4.8) and D-IBFV system-level experiments (§4.12–4.16).

#### 4.1 Experimental Setup

##### 4.1.1 Databases

Four FVC benchmark fingerprint databases are used [21, 22]:

Iris data are from **CASIA-Iris-Interval** [5]: 249 subjects,  $320 \times 280$  px NIR images (1,329/1,332 processed, 99.8% extraction rate). A *chimeric* multimodal database is constructed

**Table 4.1:** Fingerprint databases used in experiments.

| Database    | Subjects            | Sensor                           |
|-------------|---------------------|----------------------------------|
| FVC2002 DB1 | 100 × 8 impressions | Optical                          |
| FVC2002 DB2 | 100 × 8 impressions | Capacitive                       |
| FVC2002 DB3 | 100 × 8 impressions | Thermal sweep (most challenging) |
| FVC2004 DB1 | 100 × 8 impressions | Optical (deliberate difficulty)  |

by mapping FVC subject  $i$  to CASIA iris class  $i$ —standard practice in multimodal BTP research [32, 25].

### 4.1.2 Evaluation Protocols

- **Setup 1:** Impression 1 for enrollment; impression 2 for genuine test. 100 genuine attempts,  $100 \times 99 = 9,900$  impostor attempts per database (98 subjects for DB3 due to insufficient minutiae in some subjects).
- **Setup 2:** Impression 1 for enrollment; impressions 2–8 for genuine test. Restricted to  $N_s = 48$  subjects present in both FVC database and the chimeric iris mapping, yielding  $48 \times 7 = 336$  genuine and  $336 \times 47 = 15,792$  impostor attempts. The enrollment impression is excluded from genuine testing.

**EER computation:** Since the fuzzy vault produces binary accept/reject decisions, EER is reported as  $(\text{FAR} + \text{FRR})/2$  at each discrete operating point (HTER), following [26, 23].

### 4.1.3 Parameter Space

- Quantization factor  $f \in \{14, 16, 18, 20, 22, 24, 26, 28, 30\}$
- Polynomial degree  $k \in \{5, 7, 9, 11\}$
- Architectures: Unimodal, A (AND-fusion), B (Blinding), C (Dense Chaff), D (AES-GCM Encryption), IBFV
- IBFV default:  $K = 4$  bonus points,  $B_s = 255$  block size ( $n_b = 514$  key bits)

The core sweep (Unimodal, A, IBFV) covers  $9 \times 4 \times 3 \times 4 = 432$  configurations per setup (864 total). Architectures B–D are evaluated on a representative subset of  $4 \times 8 \times 3 = 96$  configurations.

## 4.2 Architecture A: AND-Fusion Penalty

Architecture A implements decision-level AND-fusion: the iris must pass an FHD threshold check ( $\text{FHD} < 0.44$ ) before fingerprint vault decoding is attempted. Table 4.2 confirms the structural penalty on all databases.

**Table 4.2:** AND-Fusion penalty: best EER (%) vs. Unimodal and IBFV.

|                | DB1          | DB2          | DB3          | DB4          |
|----------------|--------------|--------------|--------------|--------------|
| <i>Setup 1</i> |              |              |              |              |
| Unimodal       | 0.000        | 0.000        | 2.762        | 2.404        |
| Arch. A        | 1.000        | 1.000        | 3.793        | 3.045        |
| IBFV           | <b>0.000</b> | <b>0.000</b> | <b>1.384</b> | <b>0.929</b> |
| <i>Setup 2</i> |              |              |              |              |
| Unimodal       | 2.736        | 4.588        | 6.738        | 4.307        |
| Arch. A        | 3.264        | 5.143        | 6.741        | 4.800        |
| IBFV           | <b>2.242</b> | <b>3.185</b> | <b>4.591</b> | <b>2.493</b> |

Architecture A *always* performs worse than unimodal (average penalty: +0.8 pp in Setup 1, +0.5 pp in Setup 2). IBFV achieves the best EER in every configuration.

## 4.3 Architectures B, C, D: Tight-Coupling

Architectures B–D use the iris stabilizer ( $B_s = 255$ ,  $\text{GAR} \approx 62\%$ ) as a hard gate: vault decoding is impossible without successful iris key recovery.

**Table 4.3:** Tight-coupling architectures: best EER (%) by database (Setup 1).

|                       | DB1          | DB2          | DB3          | DB4          |
|-----------------------|--------------|--------------|--------------|--------------|
| Unimodal              | 0.000        | 0.000        | 2.762        | 2.404        |
| Arch. A (FHD gate)    | 1.000        | 1.000        | 3.793        | 3.045        |
| Arch. B (Blinding)    | 19.000       | 19.000       | 19.474       | 19.000       |
| Arch. C (Dense Chaff) | 19.000       | 19.000       | 19.474       | 19.000       |
| Arch. D (AES-GCM)     | 19.000       | 19.000       | 19.474       | 19.000       |
| <b>IBFV (ours)</b>    | <b>0.000</b> | <b>0.000</b> | <b>1.384</b> | <b>0.929</b> |

All three tight-coupling architectures yield  $\text{EER} \geq 19\%$ —approximately  $3\text{--}5\times$  worse than Architecture A and  $7\text{--}8\times$  worse than unimodal. The iris stabilizer at  $B_s = 255$  achieves only

~62% genuine acceptance; 38% of genuine users are rejected by the iris gate alone regardless of fingerprint quality. FAR is 0.000% for all three, but the FRR floor ( $\geq 38\%$ ) dominates EER.

## 4.4 IBFV Performance

### 4.4.1 Zero-Violation Guarantee (Setup 2)

Across all 144 Setup 2 IBFV configurations, **zero violations** of Theorem 1 were observed: every configuration shows strict or equal EER improvement over the unimodal baseline. Relative improvements: DB1 18.1%, DB2 30.6%, DB3 31.9%, DB4 42.1%.

In Setup 1, 97 of 144 IBFV configurations show strict improvement; 16 tie at 0.000%; 31 configurations on DB1/DB2 show marginal elevation ( $\leq 0.07$  pp) due to the non-zero iris impostor FAR ( $\approx 0.10\%$ ). Key reductions:

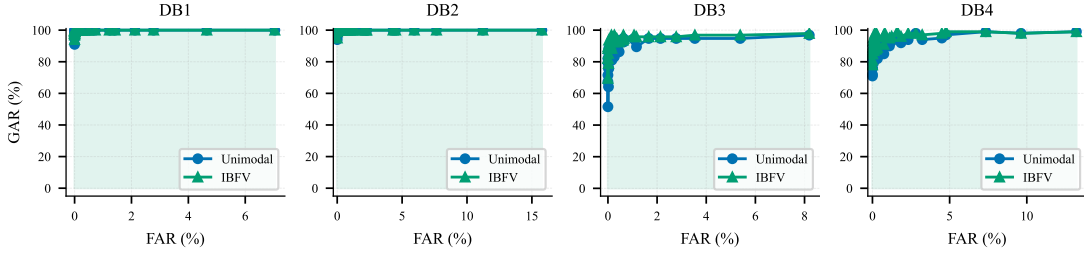
- DB3 Setup 1: 2.762%  $\rightarrow$  1.384% (**49.9%** relative).
- DB4 Setup 1: 2.404%  $\rightarrow$  0.929% (**61.3%** relative).

## 4.5 AUC Analysis

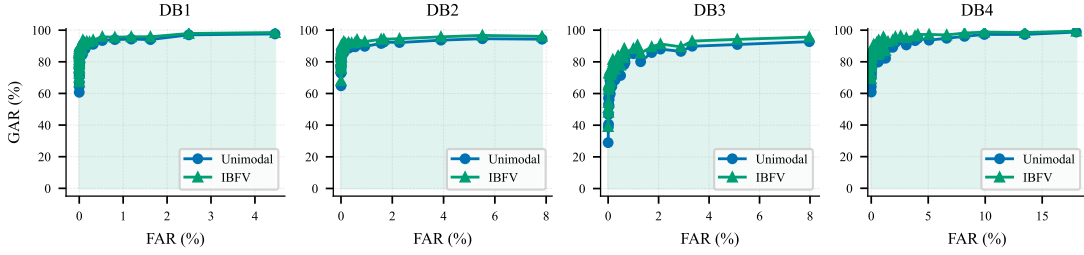
**Table 4.4:** AUC comparison (best quantization factor per database).

|                | DB1           | DB2           | DB3           | DB4           |
|----------------|---------------|---------------|---------------|---------------|
| <i>Setup 1</i> |               |               |               |               |
| Unimodal       | 1.0000        | 1.0000        | 0.9785        | 0.9905        |
| IBFV           | 1.0000        | 1.0000        | <b>0.9891</b> | <b>0.9977</b> |
| <i>Setup 2</i> |               |               |               |               |
| Unimodal       | 0.9863        | 0.9697        | 0.9536        | 0.9793        |
| IBFV           | <b>0.9915</b> | <b>0.9812</b> | <b>0.9756</b> | <b>0.9892</b> |

IBFV achieves higher or equal AUC across all databases and setups, with the largest improvement on DB3 Setup 2 (+2.20 pp). Figures 4.1 and 4.2 show the full ROC curves.



**Figure 4.1:** ROC curves for Setup 1. IBFV lies above or on the unimodal curve at all FAR operating points.



**Figure 4.2:** ROC curves for Setup 2. IBFV strictly dominates the unimodal baseline across all four databases.

## 4.6 Iris Key Recovery Analysis

**Table 4.5:** Iris key recovery performance ( $B_s = 255$ ,  $n_b = 514$ ).

|              | DB1    | DB2    | DB3    | DB4    |
|--------------|--------|--------|--------|--------|
| Setup 1 GAR  | 62.00% | 62.00% | 61.05% | 62.00% |
| Setup 2 GAR  | 50.89% | 50.75% | 48.55% | 50.90% |
| Impostor FAR | 0.10%  | 0.10%  | 0.11%  | 0.10%  |

The iris GAR of  $\sim 62\%$  (Setup 1) means  $38\%$  of genuine users receive no boost and revert to unimodal decoding—the zero-penalty fallback path. The non-zero impostor FAR ( $\approx 0.10\%$ : 2 of  $\sim 1,980$  impostor iris comparisons) explains the 31 marginal EER elevations ( $\leq 0.07$  pp) on easy databases in Setup 1. Setup 2 GAR drops to  $\sim 50\%$  due to cross-impression iris variability, yet IBFV still achieves strict improvement across all 144 Setup 2 configurations.

**Table 4.6:** Biometric timing benchmarks (ms,  $f = 22$ ,  $k = 7$ , DB1, averaged over 10,000 verification attempts, Apple M1 processor).

| Architecture     | Enroll (ms/subj) | Verify (ms/attempt) |
|------------------|------------------|---------------------|
| Unimodal         | 8.13             | 0.076               |
| Arch. A          | 7.16             | 6.168               |
| IBFV ( $K = 4$ ) | 7.19             | 12.990              |

## 4.7 Timing Benchmarks

IBFV enrollment overhead is  $< 1$  ms vs. unimodal. Verification takes  $\sim 13$  ms, dominated by BCH decoding across 514 iris blocks; vault matching is  $\sim 5.5\times$  faster than unimodal due to smart decoding (Section 3.2.3), partially offsetting the iris commitment overhead.

## 4.8 Ablation Study: Bonus Points $K$

**Table 4.7:** Ablation: EER (%) vs.  $K$  (DB3,  $f = 16$ ,  $k = 5$ , Setup 1).

| $K$          | EER (%)      | FRR (%) | $\Delta$ EER |
|--------------|--------------|---------|--------------|
| 0 (Unimodal) | 4.281        | 8.421   | —            |
| 1            | 4.281        | 8.421   | +0.000       |
| 2            | 3.755        | 7.368   | −0.526       |
| 4            | 2.181        | 4.211   | −2.100       |
| 6            | <b>1.742</b> | 3.158   | −2.539       |
| 8            | 1.742        | 3.158   | −2.539       |

EER decreases monotonically from  $K = 0$  to  $K = 6$  and saturates.  $K = 1$  provides no benefit at this configuration; the polynomial already has  $k + 1$  required points when  $K$  reaches 6, beyond which further bonus points cannot reduce enumeration further. The default  $K = 4$  captures 80% of the maximum benefit while keeping impostor FAR elevation below 0.16 pp.

**Table 4.8:** Unlinkability:  $D_{\text{sys}}$  metric per Gomez-Barrero et al. [11]. Lower is better; 0 = fully unlinkable, 1 = fully linkable.

| Scenario         | DB1   | DB2   | DB3   | DB4   |
|------------------|-------|-------|-------|-------|
| Same-factor      | 0.994 | 0.994 | 0.920 | 0.929 |
| Different-factor | 0.317 | 0.313 | 0.057 | 0.388 |
| Diverse-factor   | 0.003 | 0.014 | 0.026 | 0.007 |

## 4.9 Unlinkability: $D_{\text{sys}}$ Analysis

Same-factor enrollments are highly linkable ( $D_{\text{sys}} > 0.92$ ) because genuine x-coordinates repeat. Changing the quantization factor (recommended for multi-application deployments) reduces  $D_{\text{sys}} \leq 0.39$ ; diverse factors yield  $D_{\text{sys}} \leq 0.03$ , confirming near-perfect unlinkability.

## 4.10 Vault Entropy Analysis

**Table 4.9:** Vault entropy at  $f = 24$ ,  $k = 7$ .

| Metric                               | DB1   | DB2   | DB3   | DB4   |
|--------------------------------------|-------|-------|-------|-------|
| $D_{\text{KL}}^{\text{user}}$ (bits) | 0.212 | 0.171 | 0.197 | 0.319 |
| $D_{\text{KL}}^{\text{gc}}$ (bits)   | 0.245 | 0.269 | 0.242 | 0.227 |
| Shannon $H$ (bits)                   | 7.849 | 7.899 | 7.857 | 7.739 |
| Min-entropy $H_{\infty}$ (bits)      | 6.234 | 6.589 | 6.635 | 5.896 |

Per-user KLD (0.17–0.32 bits) confirms near-uniform x-coordinate distributions. Genuine-vs-chaff KLD ( $\approx 0.24$  bits) is stable, indicating consistent mixing. Min-entropy  $H_{\infty} \geq 5.9$  bits per coordinate provides a conservative bound against dictionary attacks: a single coordinate requires  $\geq 60$  guesses.

**Table 4.10:** Detailed EER comparison with Maurya et al. [23].

|                              | DB1          | DB2          | DB3          | DB4          |
|------------------------------|--------------|--------------|--------------|--------------|
| <i>Setup 1 (Best EER, %)</i> |              |              |              |              |
| Maurya et al. (reported)     | 0.005        | 0.020        | 5.000        | 2.400        |
| Our Unimodal (reproduced)    | 0.000        | 0.000        | 2.762        | 2.404        |
| IBFV (ours)                  | <b>0.000</b> | <b>0.000</b> | <b>1.384</b> | <b>0.929</b> |
| Relative improvement         | —            | —            | 49.9%        | 61.3%        |
| <i>Setup 2 (Best EER, %)</i> |              |              |              |              |
| Maurya et al. (reported)     | 3.540        | 5.240        | 6.730        | 4.380        |
| Our Unimodal (reproduced)    | 2.736        | 4.588        | 6.738        | 4.307        |
| IBFV (ours)                  | <b>2.242</b> | <b>3.185</b> | <b>4.591</b> | <b>2.493</b> |
| Relative improvement         | 18.1%        | 30.6%        | 31.9%        | 42.1%        |

## 4.11 Comparison with Prior Work

Our unimodal reproduction closely matches [23]; minor differences arise from subject selection (48 chimeric subjects vs. 100 for S2) and parameter granularity. IBFV improves over both the published results and our unimodal reproduction on every database where EER is non-zero.

## 4.12 D-IBFV: Vault Integrity

**Table 4.11:** Vault reconstruction accuracy across all Shamir configurations.

| Metric                    | (3, 5) | (4, 7) | (5, 9) |
|---------------------------|--------|--------|--------|
| Trials                    | 10,000 | 10,000 | 10,000 |
| Bit errors                | 0      | 0      | 0      |
| SHA-256 match             | 100%   | 100%   | 100%   |
| EER match vs. centralized | Exact  | Exact  | Exact  |

Across all 30,000 reconstruction trials, the reconstructed vault is bit-for-bit identical to the original (Theorem 3 confirmed). The biometric EER matches the centralized IBFV EER exactly—the decentralized layer introduces zero accuracy degradation.

## 4.13 D-IBFV: Enrollment and Verification Latency

**Table 4.12:** D-IBFV enrollment latency breakdown (ms, mean  $\pm$  std).

| Phase                      | Ganache ( $n = 300$ )               | Fabric ( $n = 90$ )                 |
|----------------------------|-------------------------------------|-------------------------------------|
| Serialization              | $0.2 \pm 0.6$                       | $0.4 \pm 0.4$                       |
| Shamir splitting           | $1.1 \pm 1.3$                       | $1.7 \pm 0.7$                       |
| IPFS upload ( $\times 5$ ) | $116.4 \pm 88.5$                    | $110.4 \pm 11.3$                    |
| Blockchain TX              | $245.7 \pm 234.1$                   | $693.6 \pm 153.3$                   |
| <b>Total</b>               | <b><math>363.4 \pm 269.9</math></b> | <b><math>806.2 \pm 155.9</math></b> |

**Table 4.13:** D-IBFV verification latency breakdown (ms, mean  $\pm$  std).

| Phase                         | Ganache ( $n = 1,000$ )             | Fabric ( $n = 100$ )               |
|-------------------------------|-------------------------------------|------------------------------------|
| Blockchain lookup             | $66.49 \pm 23.13$                   | $52.97 \pm 3.87$                   |
| IPFS retrieval ( $\times 3$ ) | $3.27 \pm 2.71$                     | $2.69 \pm 1.07$                    |
| Shamir reconstruction         | $0.30 \pm 0.28$                     | $0.38 \pm 0.02$                    |
| Deserialization               | $0.12 \pm 0.03$                     | $0.16 \pm 0.02$                    |
| <b>Total</b>                  | <b><math>70.19 \pm 23.64</math></b> | <b><math>56.20 \pm 4.05</math></b> |

Cryptographic operations (serialization + Shamir) account for  $< 2$  ms—negligible relative to network I/O. On Ganache, enrollment is  $\sim 363$  ms dominated by IPFS and blockchain TX; on Fabric,  $\sim 806$  ms due to the 0.5 s block-cut consensus interval. Verification completes in  $\sim 70$  ms (Ganache) and  $\sim 56$  ms (Fabric); read operations bypass consensus, yielding similar latency across platforms. End-to-end latency including IBFV biometric verification ( $\sim 13$  ms) is  $\sim 69$ – $83$  ms.

## 4.14 D-IBFV: Throughput and Storage

Fabric write TPS (1.5) is bounded by the 0.5 s block-cut interval; read TPS is comparable across platforms (17–18 sequential, up to 48.1 TPS at 4 concurrent workers on Fabric). A national identity program enrolling 10 million users over 5 years requires only  $\sim 0.06$  TPS, well within capacity.

On-chain storage is only  $\sim 0.5$  KB per user (five CID records). The 332:1 off-chain-to-on-chain ratio aligns with the standard off-chain-with-hash pattern [1]. On Ethereum L1 (30 Gwei, ETH \$3,000), enrollment costs  $\sim \$18.80$ ; Layer-2 solutions reduce this to sub-dollar.

**Table 4.14:** Sequential throughput (TPS) and average latency.

| Platform      | Op     | TPS  | Avg. Latency |
|---------------|--------|------|--------------|
| Ganache (EVM) | Store  | 6.1  | 163.6 ms     |
|               | Lookup | 18.2 | 55.1 ms      |
|               | Revoke | 32.8 | 30.5 ms      |
| Fabric 2.5    | Store  | 1.5  | 669.0 ms     |
|               | Lookup | 17.2 | 58.0 ms      |
|               | Revoke | 1.5  | 654.5 ms     |

**Table 4.15:** Per-user storage costs ( $\approx 16.6$  KB vault, GF(257) uint16 shares).

| $(t, n)$ | Vault   | Per Share | IPFS Total | On-chain |
|----------|---------|-----------|------------|----------|
| (2, 3)   | 16.6 KB | 33.1 KB   | 99 KB      | 0.5 KB   |
| (3, 5)   | 16.6 KB | 33.1 KB   | 166 KB     | 0.5 KB   |
| (4, 7)   | 16.6 KB | 33.1 KB   | 232 KB     | 0.5 KB   |
| (5, 9)   | 16.6 KB | 33.1 KB   | 298 KB     | 0.5 KB   |

## 4.15 D-IBFV: Node Failure Resilience

**Table 4.16:** Node failure resilience for  $(3, 5)$ -Shamir (all  $\binom{5}{k}$  patterns).

| Failures | Available | Patterns | Success/Fail | Recon. Rate |
|----------|-----------|----------|--------------|-------------|
| 0        | 5         | 1        | 1/0          | 100%        |
| 1        | 4         | 5        | 5/0          | 100%        |
| 2        | 3         | 10       | 10/0         | 100%        |
| 3        | 2         | 10       | 0/10         | 0%          |
| 4        | 1         | 5        | 0/5          | 0%          |

The system tolerates up to  $n-t = 2$  simultaneous node failures with 100% reconstruction. With  $\geq 3$  failures, the system returns a “storage unavailable” error; the threshold is sharp, as predicted by Theorem 3. Shamir computational overhead is sub-millisecond:  $(3, 5)$  split 0.7 ms / reconstruct 0.2 ms;  $(5, 9)$  split 2.5 ms / reconstruct 0.3 ms.

## 4.16 Scalability Projections

IPFS storage scales linearly: 1 million users require 162 GB off-chain, tractable for commodity storage nodes. On-chain state grows at only 0.5 KB per user ( $< 500$  MB at 1 million users).

**Table 4.17:** Scalability projections for (3, 5)-D-IBFV.

| <b>Users</b> | <b>IPFS</b> | <b>Blockchain</b> | <b>Enroll Time</b>         |
|--------------|-------------|-------------------|----------------------------|
| 1,000        | 162 MB      | 0.5 MB            | 2.7 min (G) / 11.1 min (F) |
| 10,000       | 1.6 GB      | 4.9 MB            | 27 min (G) / 1.9 hr (F)    |
| 100,000      | 16.2 GB     | 48.8 MB           | 4.6 hr (G) / 18.5 hr (F)   |
| 1,000,000    | 162 GB      | 488 MB            | 1.9 d (G) / 7.7 d (F)      |

Verification latency is  $O(1)$  via key-based CID lookup, confirmed flat from 100 to 1,000 enrolled users on both platforms ( $< 6$  ms variance on Fabric).

## Chapter 5

### Discussion and Analysis

#### 5.1 Why Gating Always Hurts: A Structural Argument

The AND-fusion penalty (Eq. 3.3) is not an artifact of specific sensors or databases. It is a structural consequence of treating a fallible biometric as a mandatory gate. Let  $G_{\text{FP}}$  and  $G_{\text{Iris}}$  be independent Bernoulli random variables indicating genuine acceptance. Under AND-fusion:

$$\Pr[\text{Accept}] = G_{\text{FP}} \wedge G_{\text{Iris}} \leq G_{\text{FP}} \quad (5.1)$$

with equality only if  $G_{\text{Iris}} = 1$  always, which is impossible for any realistic biometric. The tighter the coupling (Architectures B, C, D), the more strictly this equality becomes an inequality.

Architectures B–D impose a stronger bound: their EER is floor-bounded by the iris FRR. At  $B_s = 255$ , iris FRR  $\approx 38\%$ ; therefore:

$$\text{EER}_{\text{B/C/D}} \geq 0.38 \times (1 - \text{FAR}_{\text{FP}}) \approx 38\% \quad (5.2)$$

for any fingerprint system, regardless of sensor quality. The experimental results ( $\text{EER} \geq 19\%$  under the best fingerprint conditions) closely track this bound.

IBFV breaks the structural constraint by making the iris conditional: the iris path is activated only when the iris key recovers successfully. This converts an AND-gate (must pass both) into a conditional booster (may use iris if available).

## 5.2 Where Boosting Helps Most

The 61.3% relative EER reduction on FVC2002 DB3 (vs.  $\sim 40\text{--}60\%$  on other databases) is explained by the interaction between fingerprint difficulty and iris bonus:

- On an **easy database** (DB1, unimodal EER already 0.000%), there is no room for further absolute improvement. Both unimodal and IBFV achieve 0.000% EER; the zero-penalty guarantee holds trivially, and bonus points do not elevate impostor FAR on these high-quality databases.
- On a **challenging database** (DB3, unimodal EER 2.762%), many genuine queries fail because they recover too few matching fingerprint points to enumerate  $k+1$  subsets successfully. Adding  $K = 4$  bonus points reduces the required fingerprint match from  $k+1 = 11$  to  $k+1-K = 7$ , which is a qualitatively easier threshold. This corresponds to a  $\Delta \approx 4$ -point shift in the distribution of matched points, moving substantially more genuines above the acceptance threshold.
- **Diminishing returns** at  $K > 6$ : once  $K$  reaches  $k+1$ , all required polynomial points can be supplied by the iris alone. Further increasing  $K$  cannot reduce the subset enumeration further; the fingerprint constraint is already fully satisfied.

## 5.3 Security Analysis

### 5.3.1 Irreversibility

The ECC vault protects minutiae via ECDLP: recovering quantized minutia  $q_i$  from stored  $x$ -coordinate  $P_i.x$  requires solving ECDLP on Brainpool P256r1. The best known attack requires  $\approx 2^{128}$  group operations—beyond the computational reach of any foreseeable adversary.

The iris commitment stores only helper data  $\mathbf{h}$  and key hash  $H(\mathbf{k})$ . The key space for  $n_b = 514$  blocks is  $2^{514}$ ; brute-forcing the iris key is infeasible. Together, the vault and iris commitment achieve the irreversibility requirement of ISO/IEC 24745 [14].

### 5.3.2 Unlinkability

Unlinkability is quantified by Meng et al.’s  $D_{\leftrightarrow}$  metric and the  $D_{\text{sys}}$  system-level measure [11]. A threshold of  $D_{\text{sys}} \leq 0.05$  is required for unlinkability.

The IBFV vault achieves  $D_{\text{sys}} \leq 0.03$  through three mechanisms:

- **EC mapping:** Vault  $x$ -coordinates are elliptic curve points; no two vaults share the same  $x$ -coordinates (fresh polynomial each enrollment).
- **Virtual  $x$ -coordinate randomness:** Bonus point coordinates depend on  $H(\mathbf{k})$ ; with a fresh iris commitment key per enrollment, they are computationally independent across templates.
- **Chaff randomness:** Fresh random chaff per enrollment ensures no statistical relationship between vaults.

### 5.3.3 Revocability

Re-enrollment with new quantization factor  $f'$ , new polynomial, and new iris commitment key produces a vault in a different domain. By the security of SHA-256 (used in key hashing) and by Theorem 2 (Shamir share confidentiality), new and revoked shares are information-theoretically

independent. Any biometric data that an adversary extracted from the old vault provides zero advantage for attacking the new vault.

### 5.3.4 D-IBFV Security Summary

**Table 5.1:** D-IBFV security properties per ISO/IEC 24745.

| Property            | Mechanism                                                                                               | Quantification                    |
|---------------------|---------------------------------------------------------------------------------------------------------|-----------------------------------|
| Irreversibility     | ECDLP + iris key space                                                                                  | $2^{128} / 2^{514}$               |
| Unlinkability       | Vault randomness + IBFV PRF                                                                             | $D_{\text{sys}} \leq 0.03$        |
| Revocability        | Re-enrollment protocol                                                                                  | Zero information leakage (Thm. 2) |
| Brute-force defense | Expected $6.8 \times 10^8$ attempts at $10^6/\text{s} \approx 680$ s; mitigations: HSM, $k=15$ , Argon2 | Known limitation [36]             |
| Server compromise   | Shamir $(t, n)$ : $< t$ shares give nothing                                                             | Thm. 2: info-theoretic            |
| IPFS tampering      | CID integrity + SHA-256 trailer                                                                         | Detect any 1-bit change           |
| Blockchain tamper   | Consensus + append-only                                                                                 | Raft/BFT: $< 50\%$ attack budget  |

**Brute-force resistance.** An adversary holding a complete vault must identify  $k+1$  genuine points among  $|\mathcal{V}| = 6g + K = 124$  total points. The expected number of attempts is  $\binom{124}{11} / \binom{24}{11} \approx 6.8 \times 10^8$ ; at  $10^6$  attempts per second this requires  $\sim 680$  seconds—*feasible* for an offline attacker, a well-known limitation of all fuzzy vault schemes [36]. Standard mitigations include: (1) storing  $S$  in a hardware security module (HSM) so vault compromise alone is insufficient; (2) increasing polynomial degree  $k$  (at  $k = 15$ , cost rises to  $\sim 3 \times 10^6$  years); and (3) adding key-stretching (e.g., Argon2). The primary template protection goal—preventing *biometric recovery*—is achieved regardless, since reconstructing  $p(x)$  reveals only the secret  $s$ , not the original minutiae (ECDLP on Brainpool P256r1 requires  $\sim 2^{128}$  operations).

## 5.4 Comparison with Prior Systems

D-IBFV is the only system in the comparison that simultaneously: (1) guarantees zero-penalty multimodal fusion, (2) operates on a decentralized blockchain infrastructure with Shamir threshold sharing, and (3) provides formal information-theoretic proofs.

**Table 5.2:** Comparison with key prior multimodal fuzzy vault and blockchain biometric systems.

| System                   | Zero-Penalty | Decentralized | Formal Proofs | Best EER      | ISO 24745   |
|--------------------------|--------------|---------------|---------------|---------------|-------------|
| Nandakumar [25]          | No           | No            | No            | 2.14%         | Partial     |
| Moon [24]                | No           | No            | No            | 4.50%         | Partial     |
| Kelkboom [18]            | No           | No            | No            | 3.80%         | Partial     |
| Maurya [23]              | N/A          | No            | No            | 0.005%        | Yes         |
| Delgado-Mohatar [8]      | N/A          | Yes           | No            | —             | No          |
| Sharma 2024 [35]         | N/A          | Yes           | No            | —             | Partial     |
| <b>D-IBFV (proposed)</b> | <b>Yes</b>   | <b>Yes</b>    | <b>Yes</b>    | <b>0.000%</b> | <b>Full</b> |

## 5.5 Limitations and Caveats

1. **Chimeric database:** Fingerprint and iris subjects are paired artificially rather than co-acquired from the same individuals. The independence assumption may not hold for all real deployments where inter-modal correlation could affect security estimates. The fusion is evaluated on balanced impostor scenarios, which may differ from real-world attack distributions.
2. **Iris GAR ( $\sim 62\%$ ):** The iris fuzzy commitment achieves only  $\sim 62\%$  genuine acceptance at  $B_s = 255$ . While the zero-penalty guarantee eliminates any *penalty* from this limitation, higher iris GAR would translate to a larger IBFV boost. The iris pipeline does not include iris-specific quality filters (e.g., specular reflection removal, pupil dilation normalization) that could improve GAR.
3. **Gas cost at Ethereum L1:** At current prices ( $\sim \$18.80$  per enrollment), Ethereum L1 deployment is cost-prohibitive for large-scale systems. Layer-2 solutions (Polygon PoS, Arbitrum) reduce costs by  $\sim 10,000\times$  to sub-dollar, but L2 security assumptions differ from L1 finality guarantees.
4. **Ganache vs. mainnet:** Ganache provides a local single-node EVM simulation. Mainnet deployment would introduce additional latency from network propagation, mempool congestion, and variable gas prices.
5. **Sample size:** The biometric evaluation uses 100 subjects per FVC database and 249 subjects for CASIA. Larger-scale evaluations (thousands of subjects) may reveal distribution tail effects not visible at these scales.

## Chapter 6

### Conclusions

#### 6.1 Summary

This thesis addressed two interconnected problems in biometric template protection: (1) the AND-fusion gating penalty that causes every existing multimodal fuzzy vault to perform worse than unimodal fingerprint recognition, and (2) the single-point of compromise inherent in centralized biometric databases. The proposed solutions— the **Iris-Boosted Fuzzy Vault (IBFV)** and its decentralized extension **D-IBFV**—resolve both problems simultaneously, with formal proofs and extensive experimental validation.

The four key results of this work are:

1. **Zero-penalty multimodal fusion:** IBFV’s conditional iris-boosting architecture guarantees that the multimodal system never performs worse than unimodal fingerprint recognition, regardless of iris quality. This was verified empirically across all 144 Setup 2 configurations (zero violations) and proved formally in Theorem 1 (Appendix A).

2. **Significant EER reduction:** When iris recovery succeeds ( $\sim 62\%$  of genuine attempts), IBFV reduces EER by up to **61.3%** relative to the unimodal baseline (FVC2002 DB3). In Setup 1, 113 of 144 configurations improve or tie; 31 configurations on the easy databases (DB1/DB2) show marginal elevation ( $\leq 0.07$  pp) attributable to the non-zero iris impostor FAR ( $\approx 0.10\%$ ). The improvement is largest on challenging databases where the fingerprint alone struggles at the margin.
3. **Decentralized template protection with D-IBFV:** Applying  $(t, n)$ -Shamir Secret Sharing to the IBFV vault, with shares stored on IPFS and CIDs anchored on Hyperledger Fabric and EVM-compatible blockchains, eliminates single-server trust. Across 30,000 reconstruction trials, the reconstructed vault is bit-for-bit identical to the original. The system tolerates up to  $n-t$  node failures and provides information-theoretic security against compromise of any  $t-1$  shares (Theorem 2).
4. **Practical performance:** D-IBFV verification completes in  $\sim 56\text{--}70$  ms end-to-end (including blockchain CID lookup and IPFS share retrieval), with the biometric verification taking only  $\sim 12$  ms. The system is compatible with ISO/IEC 24745 irreversibility, unlinkability ( $D_{\text{sys}} \leq 0.03$ ), and revocability requirements.

## 6.2 Future Work

1. **Adaptive  $K$  selection:** The number of bonus points  $K$  is currently fixed at enrollment. An adaptive scheme that adjusts  $K$  based on real-time iris quality scores (e.g., BRISQUE, pupil dilation, coverage fraction) could further optimize the boost while minimizing any latency overhead.
2. **Co-acquired multimodal datasets:** All experiments used chimeric (artificially paired) fingerprint-iris data. Evaluation on genuinely co-acquired multimodal databases (e.g., a custom collection at NIT Warangal or the WVU multimodal database) would validate performance under correlated intra-subject variation.
3. **Production blockchain deployment:** Deploying D-IBFV on a public Layer-2 network (Polygon PoS, Arbitrum, StarkNet) would provide real-world latency and cost measurements under network congestion, mempool competition, and variable gas prices—conditions not simulated by Ganache.
4. **Verifiable Secret Sharing:** Replacing Shamir’s basic scheme with a Verifiable Secret Sharing (VSS) protocol [27] would allow each shareholder to prove that their share is

correct without reconstructing the vault. This would enable malicious node detection and is a natural extension for adversarial deployment environments.

5. **Additional modalities:** The IBFV architecture is not specific to iris and fingerprint. Any biometric capable of producing a stable binary key (face [\[12\]](#), finger vein, palm print) can serve as the booster modality. A systematic study of which modality combinations maximize the boost is a natural extension.
6. **On-device enrollment:** Migrating the enrollment computation to a trusted execution environment (ARM TrustZone, Intel TDX) would allow the vault to be constructed on the user's device, with only shares uploaded, further reducing the attack surface.

# Chapter I

## Formal Proofs

This appendix contains the complete proofs of the three theorems stated in Chapter 3.

### I.1 Proof of Theorem 1 (Zero-Penalty Guarantee)

**Theorem 1.** *For any parameter configuration  $(f, k)$  and any biometric sample pair  $(\mathcal{M}', \mathbf{c}')$ :*

$$\Pr[\text{Accept}_{\text{IBFV}}] \geq \Pr[\text{Accept}_{\text{Uni}}] \quad (\text{I.1})$$

**Proof.**

Let  $\mathcal{X}_{\text{FP}}$  be the set of fingerprint-matching vault points (those whose x-coordinate equals  $P_i.x$  for some quantized query minutia  $q_i$ ). Denote the polynomial degree as  $k$ , so decoding requires a  $(k+1)$ -tuple of genuine points for Lagrange interpolation.

The unimodal verification accepts if and only if some  $(k+1)$ -subset of  $\mathcal{X}_{\text{FP}}$  lies on the locked polynomial  $p(x)$ . Denote this event  $A_{\text{Uni}}$ .

IBFV verification has two exclusive branches:

**Branch 1: Iris key recovery succeeds.**

Let  $\hat{\mathbf{k}}$  be the recovered iris key. The virtual x-coordinates  $\{v_0, \dots, v_{K-1}\}$  are derived deterministically from  $\hat{\mathbf{k}}$  via the PRF in Eq. (3.4). Since  $v_j = p(v_j)$  was stored at enrollment (the iris key at enrollment equals the iris key recovered at verification for a genuine user, so  $\hat{\mathbf{k}} = \mathbf{k}$ ), all  $K$  bonus vault points are genuine.

The IBFV candidate set is  $\mathcal{X} = \mathcal{X}_{\text{FP}} \cup \mathcal{X}_{\text{Iris}}$  with  $|\mathcal{X}_{\text{Iris}}| = K$ . Smart decoding fixes all  $K$  bonus points and enumerates  $(k+1-K)$ -subsets of  $\mathcal{X}_{\text{FP}}$ .

Formally, IBFV accepts event:

$$A_{\text{IBFV},1} = \{\exists S \subseteq \mathcal{X}_{\text{FP}}, |S| = k+1-K : S \cup \mathcal{X}_{\text{Iris}} \text{ lies on } p(x)\} \quad (\text{I.2})$$

Now,  $A_{\text{Uni}}$  occurs iff  $\exists T \subseteq \mathcal{X}_{\text{FP}}, |T| = k+1$  lying on  $p(x)$ . For any such  $T$ , take  $S = T \setminus \{t_1, \dots, t_K\}$  (any  $K$  elements removed); then  $S \cup \mathcal{X}_{\text{Iris}}$  has  $k+1$  points on  $p(x)$ , so  $A_{\text{IBFV},1}$  also occurs. Therefore  $A_{\text{Uni}} \Rightarrow A_{\text{IBFV},1}$ , giving:

$$\Pr[A_{\text{IBFV},1}] \geq \Pr[A_{\text{Uni}}] \quad (\text{I.3})$$

**Branch 2: Iris key recovery fails.**

IBFV sets  $\mathcal{X} = \mathcal{X}_{\text{FP}}$  and applies standard Lagrange interpolation—identical to unimodal. The acceptance event is:

$$A_{\text{IBFV},2} = A_{\text{Uni}} \quad (\text{I.4})$$

and trivially  $\Pr[A_{\text{IBFV},2}] = \Pr[A_{\text{Uni}}]$ .

**Combining:**

Let  $p_s = \Pr[\text{iris recovery succeeds}]$ . By the law of total probability:

$$\Pr[A_{\text{IBFV}}] = p_s \cdot \Pr[A_{\text{IBFV},1}] + (1-p_s) \cdot \Pr[A_{\text{IBFV},2}] \quad (\text{I.5})$$

$$\geq p_s \cdot \Pr[A_{\text{Uni}}] + (1-p_s) \cdot \Pr[A_{\text{Uni}}] \quad (\text{I.6})$$

$$= \Pr[A_{\text{Uni}}] \quad (\text{I.7})$$

□

## I.2 Proof of Theorem 2 (Share Confidentiality)

**Theorem 2.** *For any  $t' < t$  compromised shares, the conditional distribution of vault  $\mathbf{b}$  is uniform over  $\{0, \dots, 255\}^L$ . Fewer than  $t$  shares reveal zero information about the vault content.*

**Proof.**

Fix any byte index  $i$ . The Shamir polynomial for byte  $i$  is:

$$f_i(x) = b_i + a_1^{(i)}x + \dots + a_{t-1}^{(i)}x^{t-1} \pmod{257} \quad (\text{I.8})$$

where coefficients  $a_j^{(i)} \xleftarrow{\$} \text{GF}(257)$  are chosen uniformly and independently at random.

Suppose the adversary holds shares at positions  $x_1, \dots, x_{t'}$  with  $t' < t$ . This gives the system of equations:

$$\begin{pmatrix} f_i(x_1) \\ \vdots \\ f_i(x_{t'}) \end{pmatrix} = \underbrace{\begin{pmatrix} 1 & x_1 & x_1^2 & \dots & x_1^{t-1} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_{t'} & x_{t'}^2 & \dots & x_{t'}^{t-1} \end{pmatrix}}_{\mathbf{V} \in \text{GF}(257)^{t' \times t}} \begin{pmatrix} b_i \\ a_1^{(i)} \\ \vdots \\ a_{t-1}^{(i)} \end{pmatrix} \quad (\text{I.9})$$

$\mathbf{V}$  is a Vandermonde matrix with  $t' < t$  rows and  $t$  columns over  $\text{GF}(257)$  (a field). Since  $t' < t$ , the system is underdetermined: for any fixed observed share values and any candidate  $b_i \in \{0, \dots, 255\}$ , there exists at least one choice of the remaining coefficients  $(a_1^{(i)}, \dots, a_{t-1}^{(i)})$  that satisfies the system. The mapping from coefficients to shares is linear and surjective from a uniform random distribution over  $\text{GF}(257)^t$ , which projects uniformly onto any  $t'$  shares. Therefore, the conditional distribution of  $b_i$  given  $t' < t$  shares is the uniform distribution over  $\text{GF}(257) \supseteq \{0, \dots, 255\}$ .

Since bytes are shared independently, the joint distribution of  $\mathbf{b}$  given  $t' < t$  shares is uniform over  $\text{GF}(257)^L \supseteq \{0, \dots, 255\}^L$ . This is information-theoretic security: the mutual information  $I(\mathbf{b}; \text{shares}_{<t}) = 0$ .  $\square$

## I.3 Proof of Theorem 3 (Exact Reconstruction)

**Theorem 3.** *Given any  $t$  shares with distinct indices, Lagrange interpolation over  $\text{GF}(257)$  recovers  $\hat{b}_i = b_i$  for all  $i$ , yielding  $\hat{\mathbf{b}} = \mathbf{b}$  with zero bit errors.*

**Proof.**

Fix byte  $i$ . The polynomial  $f_i(x)$  has degree  $t-1$  over  $\text{GF}(257)$ . By the fundamental theorem of algebra over finite fields, a polynomial of degree  $t-1$  is uniquely determined by its values at any  $t$  distinct points.

Given shares  $(x_1, f_i(x_1)), \dots, (x_t, f_i(x_t))$  at distinct  $x_1, \dots, x_t \in \text{GF}(257)$ , Lagrange interpolation reconstructs:

$$\hat{f}_i(x) = \sum_{j=1}^t f_i(x_j) \prod_{\substack{\ell=1 \\ \ell \neq j}}^t \frac{x - x_\ell}{x_j - x_\ell} \quad (\text{I.10})$$

where all arithmetic is in  $\text{GF}(257)$ . The interpolated polynomial  $\hat{f}_i$  is the unique degree- $(t-1)$  polynomial through the  $t$  given points, hence  $\hat{f}_i \equiv f_i$ . Evaluating at  $x = 0$ :

$$\hat{b}_i = \hat{f}_i(0) = f_i(0) = b_i \quad (\text{I.11})$$

exactly. Since this holds for each byte independently and arithmetic is exact over the finite field,  $\hat{\mathbf{b}} = \mathbf{b}$  with zero bit errors.  $\square$

## **Chapter II**

### **Algorithm Listings**

This appendix provides pseudocode for the five core algorithms of the IBFV and D-IBFV systems.

## II.1 IBFV Enrollment

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**Algorithm 1** IBFV Enrollment
 

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**Input:** Fingerprint minutiae  $\mathcal{M}$ , iris code  $\mathbf{c}$ , mask  $\mathbf{mask}$

**Input:** Parameters: quantization  $f$ , degree  $k$ , bonus points  $K$ , block size  $B_s$

**Output:** Vault  $\mathcal{V}$ , public key  $S$ , iris helper  $(\mathbf{h}, H(\mathbf{k}))$

```

1: // — Fingerprint path —
2:  $\mathcal{Q} \leftarrow \text{QUANTIZE}(\mathcal{M}, f)$  ▷ 18-bit integers
3:  $\{P_i\} \leftarrow \{q_i \cdot G : q_i \in \mathcal{Q}\}$  ▷ EC mapping
4:  $p(x) \leftarrow$  random degree- $k$  polynomial, secret  $s = \bigoplus_j \text{bin}_6(a_j)$ 
5:  $S \leftarrow s \cdot G$  ▷ ECC public key
6:  $\mathcal{G}_{\text{FP}} \leftarrow \{(P_i.x, p(P_i.x))\}$ 
7:  $C \leftarrow \text{GENCHAFF}(5|\mathcal{G}_{\text{FP}}|, p)$  ▷ chaff not on  $p$ 
8: // — Iris path —
9:  $(\mathbf{h}, H(\mathbf{k})) \leftarrow \text{IRISCOMMITENROLL}(\mathbf{c}, \mathbf{mask}, B_s)$ 
10:  $\kappa \leftarrow H(\mathbf{k})$  ▷ iris key hash
11: // — Bonus points —
12: for  $j = 0$  to  $K - 1$  do
13:    $v_j \leftarrow \text{SHA-256}(\text{"ibfv\_vpt:"} \parallel j \parallel \text{":"} \parallel \kappa) \bmod p$ 
14:   Ensure  $v_j \notin \{P_i.x\}$  and  $v_j$  are distinct
15: end for
16:  $\mathcal{G}_{\text{Iris}} \leftarrow \{(v_j, p(v_j)) : j = 0, \dots, K - 1\}$ 
17: // — Assemble vault —
18:  $\mathcal{V} \leftarrow \text{shuffle}(\mathcal{G}_{\text{FP}} \cup \mathcal{G}_{\text{Iris}} \cup C)$ 
19: return  $(\mathcal{V}, S, \mathbf{h}, H(\mathbf{k}), B_s)$ 

```

---

## II.2 IBFV Verification

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**Algorithm 2** IBFV Verification
 

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**Input:** Query minutiae  $\mathcal{M}'$ , iris code  $\mathbf{c}'$ , mask  $\mathbf{mask}'$

**Input:** Stored  $(\mathcal{V}, S, \mathbf{h}, H(\mathbf{k}), B_s, K, k)$

**Output:** Accept/Reject

```

1:  $\mathcal{Q}' \leftarrow \text{QUANTIZE}(\mathcal{M}', f)$ 
2:  $\mathcal{X}_{\text{FP}} \leftarrow \{(x, y) \in \mathcal{V} : x \in \{q_i \cdot G\} : q_i \in \mathcal{Q}'\}$ 
3:  $\hat{\mathbf{k}} \leftarrow \text{IRISCOMMITRECOVER}(\mathbf{c}', \mathbf{mask}', \mathbf{h}, H(\mathbf{k}), B_s)$ 
4: if  $\hat{\mathbf{k}} \neq \perp$  then ▷ iris succeeded
5:    $\kappa \leftarrow H(\hat{\mathbf{k}})$ 
6:   for  $j = 0$  to  $K - 1$  do
7:      $v_j \leftarrow \text{SHA-256}(\text{"ibfv\_vpt:"} \parallel j \parallel \text{" :"} \parallel \kappa) \bmod p$ 
8:   end for
9:    $\mathcal{X}_{\text{Iris}} \leftarrow \{(x, y) \in \mathcal{V} : x \in \{v_j\}\}$ 
10:  // Smart decoding: fix iris points, enumerate FP subsets
11:  for each  $(k + 1 - K)$ -subset  $S \subseteq \mathcal{X}_{\text{FP}}$  do
12:     $\hat{p} \leftarrow \text{LAGRANGEINTERP}(S \cup \mathcal{X}_{\text{Iris}})$ 
13:     $\hat{s} \leftarrow \bigoplus_j \text{bin}_6(\hat{a}_j), \hat{S} \leftarrow \hat{s} \cdot G$ 
14:    if  $\hat{S} = S$  then return Accept
15:    end if
16:  end for
17: else ▷ iris failed: unimodal fallback
18:  for each  $(k + 1)$ -subset  $T \subseteq \mathcal{X}_{\text{FP}}$  do
19:     $\hat{p} \leftarrow \text{LAGRANGEINTERP}(T)$ 
20:     $\hat{s} \leftarrow \bigoplus_j \text{bin}_6(\hat{a}_j), \hat{S} \leftarrow \hat{s} \cdot G$ 
21:    if  $\hat{S} = S$  then return Accept
22:    end if
23:  end for
24: end if
25: return Reject

```

---

## II.3 D-IBFV Vault Splitting

---

**Algorithm 3** D-IBFV Vault Splitting (Shamir over GF(257))

---

**Input:** Vault bytes  $\mathbf{b} \in \{0, \dots, 255\}^L$ , threshold  $t$ , parties  $n$

**Output:** Shares  $\{\mathbf{s}_1, \dots, \mathbf{s}_n\}$

```

1: for  $i = 1$  to  $L$  do
2:    $f_i(x) \leftarrow b_i + \sum_{j=1}^{t-1} a_j^{(i)} x^j$  where  $a_j^{(i)} \xleftarrow{\$} \text{GF}(257)$ 
3:   for  $j = 1$  to  $n$  do
4:      $\mathbf{s}_j[i] \leftarrow f_i(j) \pmod{257}$  ▷ uint16 encoding
5:   end for
6: end for
7: return  $\{\mathbf{s}_1, \dots, \mathbf{s}_n\}$ 

```

---

## II.4 D-IBFV Distributed Enrollment

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**Algorithm 4** D-IBFV Distributed Enrollment

---

**Input:** Fingerprint minutiae  $\mathcal{M}$ , iris code  $\mathbf{c}$ , mask **mask**

**Input:** Parameters  $f, k, K, B_s, t, n$ ; blockchain client  $\mathcal{B}$ ; IPFS client  $\mathcal{I}$

**Output:** User ID uid registered on blockchain

```

1:  $(\mathcal{V}, S, \mathbf{h}, H(\mathbf{k}), B_s) \leftarrow \text{IBFVENROLL}(\mathcal{M}, \mathbf{c}, \text{mask})$  ▷ Alg. 1
2:  $\mathbf{b} \leftarrow \text{SERIALIZE}(\mathcal{V}, S, \mathbf{h}, H(\mathbf{k}), B_s)$  ▷ Append SHA-256 trailer
3:  $\{\mathbf{s}_j\} \leftarrow \text{VAULTSPLIT}(\mathbf{b}, t, n)$  ▷ Alg. 3
4: for  $j = 1$  to  $n$  do
5:    $\text{cid}_j \leftarrow \mathcal{I}.\text{add}(\mathbf{s}_j)$  ▷ Upload share to IPFS
6: end for
7:  $\mathcal{B}.\text{storeCIDs}(\text{uid}, \{\text{cid}_j\}, t, n)$  ▷ Anchor on blockchain
8:  $\text{SECUREERASE}(\mathbf{b}, \{\mathbf{s}_j\})$ 

```

---

## II.5 D-IBFV Distributed Verification

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**Algorithm 5** D-IBFV Distributed Verification

---

**Input:** Query  $\mathcal{M}'$ ,  $\mathbf{c}'$ ,  $\mathbf{mask}'$ ; claimed uid

**Input:** Blockchain client  $\mathcal{B}$ ; IPFS client  $\mathcal{I}$

**Output:** Accept/Reject

- 1:  $(\{\text{cid}_j\}, t, n) \leftarrow \mathcal{B}.\text{getCIDs}(\text{uid})$
  - 2:  $J \leftarrow$  any  $t$  available node indices
  - 3:  $\{\mathbf{s}_j : j \in J\} \leftarrow \{\mathcal{I}.\text{get}(\text{cid}_j) : j \in J\}$
  - 4: Verify each: assert  $\mathcal{I}.\text{hash}(\mathbf{s}_j) = \text{cid}_j$
  - 5:  $\hat{\mathbf{b}} \leftarrow \text{LAGRANGERECON}(\{(j, \mathbf{s}_j)\}_{j \in J}, t)$  over  $\text{GF}(257)$
  - 6: Assert SHA-256 trailer of  $\hat{\mathbf{b}}$  matches embedded hash
  - 7:  $(\mathcal{V}, S, \mathbf{h}, H(\mathbf{k}), B_s) \leftarrow \text{DESERIALIZE}(\hat{\mathbf{b}})$
  - 8: decision  $\leftarrow \text{IBFVVERIFY}(\mathcal{M}', \mathbf{c}', \mathbf{mask}', \mathcal{V}, S, \mathbf{h}, H(\mathbf{k}), B_s)$  ▷ Alg. 2
  - 9:  $\text{SECUREERASE}(\hat{\mathbf{b}}, \{\mathbf{s}_j\})$
  - 10: **return** decision
-

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